

THE SCIENCE OF METHODS

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Register of the Elements of Method

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To my teachers of formal systems:

MOHAN PUTCHA
for the Semigroups

GERARD HOCHSCHILD
for the Descent

SHIING-SHEN CHERN
for the Spherical Manifold

Contents

PART I — FOUNDATIONS	1
PART II — GRAMMAR OF METHOD	15
PART III — OPERATIVE MODELS AND TRANSPOSITIONS	25
PART IV — MACHINES	43
PART V — SYSTEMS OF METHODS AND METHOD SPACE	49
PART VI — METAPHOR	55
ANNEX 0 — AXIOMATICS AND ALGEBRA OF METHODICITY	59
ANNEX I — FUNCTORIAL AND SHEAF- THEORETIC INTEGRATION	69
ANNEX II — N-TUPLE MATRIX	79
ANNEX III — MATRIX OF CONFIGURATION SPACES	89
APPENDIX A — THE ATLAS	101
APPENDIX B — TESTING	109
APPENDIX C — SYSTEMIC CASE STUDY	121

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Eric Cafritz (2026)

This work examines method itself. A method is a system with identifiable parts and regularities. The Register develops an axiomatic treatment of that system. Detailed formal restatements are contained in annexes to preserve continuity of exposition.

PART I — FOUNDATIONS

A method starts with the arrangement of a field. Relations appear when the field is divided into regions that can be recognized and compared. Meaning arises from these relations. The initial steps of method are the establishment of a field of inquiry and the decision to treat designated relations as objects of study.

Foundations describe the formative conditions of method.

Core Principle

Human thought lies at the center of the Science of Methods. Methods are interior structures of attention, selection, relation, and expression. Human thought shapes method, and method in turn shapes thought.

While method in a strict sense is a later cultural innovation, its roots lie in the evolutionary advantages of attention to sequence, memory of cause and effect, imitation, and incremental correction.

Human society itself is shaped by method. Long before formal systems of any kind, groups sustained their life by inventing repeatable ways of acting together through recognition, exchange, care, conflict, and coordination. These acts are essential acts of method.

Foundational Axiom

Meaning arises from the arrangement of relations.

Meaning appears when relations are arranged in a way that makes a pattern perceptible. A change in arrangement produces a change in meaning even when the elements remain the same. Arrangement is the opening act of intelligibility.

Objects as Relations

An object, as it appears in a domain of inquiry, is already a relation. It stands out from what surrounds it and is distinguished from the rest of the domain. This distinction is a difference, and in the Register a difference is a relation.

An object is a segment of the domain that the observer has designated or recognized as distinct. Even the first object in a domain comes into being only because the observer has drawn a boundary between 'this' and 'not this'. That boundary is a relation.

An object is a relation viewed as an element of inquiry, available for arrangement, examination, and combination with other relations.

Configurations and Steps

A configuration is the present arrangement of the relations the practitioner has designated, ordered, or set aside for further study.

Method proceeds by steps. A step is a deliberate act by which the practitioner alters the present configuration of designated relations in response to one or more selected relations. The algebraic representation of this progression, treating steps as generators of a sequence, is established in Annex 0.

In the Register, the term 'structure' has a specific meaning derived from the methodic primitives of relations and steps: 'structure' describes a configuration designated for continuation.

Agency

Meaning arises from the arrangement of relations, but method requires the act of an agent to identify arrangements for study.

Agency is a property of the one who enacts the method. When the Register describes a method as adapting or revising, the expression is shorthand for changes introduced by an agent working through it, or by rules the agent has embedded within its operation. Even in self-modifying or recursive procedures reconfiguration follows from design by an agent.

The early phases of inquiry depend heavily on deliberate agency. Attention must be actively directed, distinctions intentionally drawn, and relations tentatively explored. At this stage, coherence is fragile, and progress relies on sustained intervention by an observer who selects, tests, and revises provisional arrangements.

As coherence builds, the role of agency changes. The designation of relations for study constrains the next steps, narrowing subsequent action. What once required active choice becomes increasingly guided by the established configuration. Less effort is devoted to discovering relations, and more to recognizing and respecting those that have already proven coherent. This shift marks a threshold: beyond it, inquiry proceeds less by open-ended construction and more by following implications from prior work.

Eventuation

Method starts with an external impulse that causes the observer to commit attention to a domain. The cause or origin of this impulse lies outside method itself, commonly in necessity or curiosity. A sequence of steps follows and concludes in an outcome that may succeed, fail, or require revision.

This eventive character is constitutive. Steps are taken, irreversibly; coherence accumulates; and revision becomes possible, all in temporal order. Because methods are played out

in actual time, they are exposed to contingency, error, and exhaustion. Their efficacy depends on the configuration of steps, and on duration, sequence, and termination.

Higher formal disciplines, such as mathematics and logic, depend on method. Yet among these method is distinguished by its essential eventuation. Mathematics and logic can be defined, examined, and evaluated without reference to their performance in time; methods cannot be understood fully apart from their enactment.

Method is capable of transmutation. The steps of inquiry may produce abstract objects, such as organizational models, which themselves can become objects of inquiry. These objects may not have existed prior to the inquiry, and they are not necessarily inherent in the field initially examined.

Pre-Inquiry: Purpose and Intent

At the outset, there is only a direction held in awareness, with a sense of what the work will address and a commitment to begin.

Purpose establishes the direction of inquiry. Intent sustains the selection of steps in accordance with purpose. These movements occur together. As steps accumulate, orientation indicates whether the direction set by purpose has been maintained.

When purpose is absent, the work drifts. When intent is absent, the work collapses for lack of engagement.

Designation, Constitution, Naming

A field must be divided before method can begin. Designation determines what will count as an object; constitution is the act of placing designated objects in an arrangement for study; naming fixes a designated object so it can be referred to,

compared, and used in arrangements.

An object of study is rarely simply found. It is selected through deliberate choice, which joins description with constitution.

The decision to treat something as an object already implies an interpretation of its role in the inquiry. That decision reflects attention and purpose.

Fuzziness and Uncertainty

Fuzziness is the condition in which differences are present but not yet sharply distinguished. The practitioner discerns differences within a domain, but their separation is incomplete: boundaries overlap, distinctions are partial, and relations among differences are obscured. A greater degree of fuzziness corresponds to a wider diffusion among differences, where it remains unclear how one difference is separated from another or how several differences relate among themselves. As method proceeds, fuzziness may be reduced as differences are more clearly distinguished and configurations are designated for further steps.

Uncertainty refers to a state in which differences and relations have been discerned and arranged into a configuration, but the consequences of acting on that configuration are not yet settled. The practitioner has designated a present arrangement of relations, but cannot yet determine how further steps will alter the field or whether those steps will further the intended purpose. Uncertainty results from the limits of foresight, the incompleteness of available relations, and the fact that method proceeds through time. As steps are taken and configurations are altered, uncertainty may be reduced or displaced, but it remains intrinsic to method wherever continuation occurs without full determination.

Veracity and Plausibility

A configuration reflects an accumulation through steps of directed attention and selection based on purpose. Veracity is the extent to which a configuration remains oriented towards purpose. Veracity may have to be recalibrated if purpose shifts and can only be finally assessed, if ever, after completion of the application.

Method progresses through steps; plausibility governs the decision to take them. It operates where veracity is not yet settled.

A configuration is plausible when further steps can continue it in a way that the practitioner determines will likely, but not necessarily, preserve orientation towards purpose. Plausibility is a gamble. If the gamble fails, the method drifts.

Coherence

Coherence is the sufficiency of a configuration for continuation, as determined by the practitioner. This determination is subject to the interplay of four variables: veracity, alignment, intensity, and uncertainty.

Veracity and uncertainty set the limiting contours of coherence. Alignment measures fit and orientation among steps within a configuration; it is the geometric dimension of order. Veracity, alignment, and uncertainty determine whether the configuration can be maintained and carried forward. Intensity reflects amplitude: the degree to which a configuration is prominent.

The determination is local, based on the state of the configuration at the time the next step is taken. If the determination is in error, the method drifts. Drift is the result of error, whether of prediction (plausibility) or assessment (coherence).

Plausibility operates by intimation; coherence bears the weight of an assessment under the four variables. The difference between the two is often indistinct. It is resolved only where the determination is obvious, dictated by local circumstance, or revealed by hindsight.

Accident as Information

Accident appears by accident: it is an unplanned deviation, but it may reveal a relation or a contradiction not previously recognized. When treated as information, accident can alter orientation.

Accidents generate differences not produced by intention or internal operations. Once encountered, these differences are subject to selection: some may be incorporated into the method, while others are discarded. The novelty introduced is external; it arrives from the field rather than from within the method itself, which remains an open system.

Consciousness

Consciousness provides the initial ground of method. Conscious awareness opens a mental space in which objects, relations, and distinctions can be held long enough to reveal significance and meaning. In this mental space, contrast, form, and context can be organized and shifted in relation to one another.

The operative gesture of consciousness is attention. Attention identifies what stands out in the field of inquiry, giving an early contour of a problem. Much cognitive activity proceeds without awareness, but a deliberate method begins with conscious attention.

Within consciousness, a form can be maintained long

enough for comparison with what appears, revealing divergence. This is an error signal, indicating a need for adjustment.

Method relies on representations, such as signs, symbols, and beacons. A symbol carries sense because consciousness can link sign and referent. A beacon can shift the focus of conscious viewing while the object of study remains steady.

Consciousness is reflexive. It can turn toward its own operations, making movements of attention, acts of selection, and underlying assumptions available for inspection.

Pattern Recognition

Neuroscience uses the term ‘pattern processing’ to describe the brain’s capacity to register and relate patterns in perception, memory, and imagination. As the human brain evolved, its capacity to recognize and work with patterns improved, allowing them to be formed, tested, and shared in deliberate ways. Method depends on this capacity.

Mediation

Even the simplest methods proceed through steps held initially in thought and only then applied to a particular field. In many cases, such as simple measurement by hand, the field responds immediately, and this mental mediation goes unnoticed. In others, such as complex games like Go or structured operations like surgery, the configuration must be carried in thought and revised internally over many steps before and during application to the field.

The first relations are arranged in the mind, before the field is engaged.

Taxonomy

Classification is essential to living science. The discernment of kind, order, and relation precedes experiment and proof.

Linnaeus, the father of modern taxonomy, arranged the profusion of living organisms into visible categories of resemblance. The Register compares principles and operations of understanding and arranges them into a morphology of method.

Each component, technique, and sign of method is a morphological element of thought.

Linnaeus's work depended on attunement to proportion and pattern. Method depends on a similar attunement, a sensitivity to the relations that constitute intelligibility.

Evolution of Methods

In practice, methods are adapted to their conditions of use. Changes in the field introduce new relations that may require attention. A step that once fit under one set of conditions may require a different placement when the field, aims, or configuration changes. These shifts affect how the configuration should be set.

Small variations often enter through error, improvisation, or unusual conditions. Variations that maintain coherence tend to recur, and those that fail to do so tend to fall away.

Different methods often exist for the same task, and their performance can be compared. Some may be simpler, faster, or more productive in implementation than others. A qualitative comparison rests on how well each one serves the purpose defined by that task.

When a method yields reliable results, it tends to be used again. Repetition stabilizes steps, making execution easier and increasing the likelihood of transmission.

These patterns of differentiation, recurrence, persistence, and disappearance follow principles recognizably close to those of evolution.

Darwin as Observational Method

Darwin reinterpreted the hierarchy of Linnaean taxonomy as historical: he argued that living beings fall into nested groups because they share common ancestry, where Linnaeus had treated these groups as ahistorical, static categories. Yet Darwin did not begin with an abstract theory. He worked, almost modestly, with field notes: lists of specimens, sketches of morphology, comparisons of habits and habitats, brief remarks on distribution and variation. These accumulations of small observations, drawn especially from the Beagle voyage and his later notebooks, gradually revealed patterns of divergence and continuity that no fixed-species view could account for. A method crystallized only afterward.

As Darwin's observations multiplied, relations became clearer, contrasts took on significance, and a system of explanatory connections formed. His method developed from the observational practice itself, as opposed to prior prescription. Through careful, sustained attention, Darwin arrived at his own method, an evolution eerily echoing his later theory.

Practical Origins

Trial and Error

Inquiry often proceeds by trial and error: an attempt, a consequence, and an adjustment that modifies the next attempt. This cycle precedes explicit measure or rule. It is a behavioral pattern in which action and correction take recognizable form.

Measure

Measure is a way of relating one span, interval, or quantity to another. A hand-span pressed into soil to space seeds in a row, a rope pulled taut between two stakes, a stick dipped to gauge depth, or a counted series of paces over open ground are elementary expressions of proportion. These gestures allow the practitioner to repeat a result, record a difference, or hold a boundary.

Measure is a transposition that translates the physical world into a sequence of comparable relations. A unit is chosen and applied to a field, which transfers an unknown quantity and relates it back to something already familiar.

Once a measure exists, operations follow. A sequence of marks leads to counting. Repetition foreshadows estimation. Alignment and composition of measurements sketch layout and form. Measure is an inquisitive act that one can compose, record, and return to.

More elaborate methods retain these qualities. Calibration, scaling, and verification are refinements of measurement. The regularity of measure precedes calculation but already points to formal correspondence.

Performance

Some methods are enacted directly through performance: ritual, experiment, music, theater. In these cases, the act itself provides an orientation towards the outside world.

But even a practical 'how-to' method starts in inquiry: recipes and other craft techniques are derived from trial-and-error; engineering procedures use iterative analysis; and algorithms are built through stepwise refinement. In the course of inquiry, the practitioner examines a domain, identifies relevant

relations, tests arrangements, corrects failures, and refines the sequence of actions that produce a desired result. Once refined, those actions can be carried out without further examination.

Performative methods result from an earlier inquiry the internal steps of which have settled into a reliable and repeatable sequence of actions. The line between inquiry and doing is gradual.

Utility

Many methods are devised to solve practical situations. They originate as a way to build, calculate, decide, or even to survive.

Horticulture is one of the earliest identifiable examples of organized methodical action. Gardeners discovered through practice the conditions that later become explicit in formal method.

The actions of horticulture express methodical relations. Soil is prepared according to texture, moisture, and season. Seeds are placed at a depth suited to their type. Watering follows interval, proportion, and climate. Each plant reveals its own requirements for spacing, light, or pruning. Successful gardening requires fit between act and environment, veracity as responsiveness to changing conditions, and reproducibility to ensure the return of growth.

Horticulture depends on observation, patience, correction, and a disciplined refusal to rush the field. The gardener learns arrangement by attending to the layout of beds and rows; measure by using hand, depth, or interval to fix amounts or intervals; attunement by sensing when a plant is stressed; and accident as information when weather or pests indicate the need for adjustment. This practice displays a full cycle of methodic action: attention, arrangement, inference, and evaluation appear in each season's work.

Art

Art shows how method behaves in a field without fixed steps. Artistic work often takes shape gradually, in the act of making. A painter adjusts a line because it looks too heavy. A musician alters a phrase because it sounds unbalanced. These adjustments follow patterns that experienced artists recognize, even when the patterns are not written down as rules.

Art is usually meant to convey meaning, not to be practically useful, but it can serve as a means of inquiry in fields far removed from art. A child who is asked to draw during a psychological evaluation may express concerns, ideas, or relations through marks on the page. These marks show emphasis, repetition, and avoidance. The drawing is a field with enough definition to reveal what matters to the child. It is an organized action that surfaces tensions or themes that might not appear in response to questions alone.

Some artistic processes resemble adaptive methods. An artist may begin with a plan, but the plan changes as the work proceeds. The change responds to the demands of the material and the proportions that appear. The artist keeps the work coherent by responding to these conditions. In this adaptive mode, the sequence of steps is determined by the developing configuration rather than a fixed plan.

PART II — GRAMMAR OF METHOD

In a methodical process, a practitioner employs a repertoire of gestures and tools, which are elemental operations of inquiry.

The grammar is a morphology: a system of components, techniques, and evaluative criteria that constitute the repertoire of method.

Components

Components are the operative gestures by which a field of inquiry is differentiated. They make distinctions available for further organization, transformation, and evaluation.

Perceptual components clarify what is available for inquiry. Observation opens a field. Noticing and detection register features or changes. Discrimination separates what might otherwise remain undifferentiated. Examination, scrutiny, and inspection increase resolution when finer distinctions are required.

Provisional components include suggestion, guessing, approximation, and trial configuration. These gestures permit inquiry to begin and continue in the absence of full justification. They generate proposals, orientations, and provisional approaches that can later be corrected or confirmed, providing early sources of concepts and direction.

Constructive components operate by dividing, grouping, and arranging what has been distinguished in a field of inquiry. They include separation, partitioning, juxtaposition, assembling, and more broadly, organization. Construction itself belongs in this group, in the sense of a rule-governed production of a configuration through an explicit sequence of permitted operations, as in geometric or algorithmic construction.

Transformational components include rotation, reduction, expansion, translation, projection, and transposition. These modify a configuration by changing frame, scale, or representation, or by moving a configuration into a different space where other relations become visible.

Regulatory and evaluative components include testing and qualitative assessment. These gestures examine a configuration for adequacy and incompatibility, serving to determine whether further operations are warranted or whether the inquiry has reached a point of closure.

Symbols and expressions, such as labels and formulas, condense relations into usable proxies. They allow configurations to be encapsulated, combined, and compared without having to reconstruct every underlying step.

Techniques

A technique is a repeatable way of combining or refining components into a procedure that can be reapplied and varied. Techniques differ by the role they play in the development of an inquiry.

In their simplest form, techniques are a set of procedural moves that assist in local analysis. They can be used like tools in a kit and can be stitched together, swapped in sequence, and repeated.

Selection and filtering restrict a field by limiting what enters it, which focuses scrutiny. Dissection and organizing make relations explicit by separating, ordering, or grouping material. Recording preserves configurations so they can be revisited, compared, or tested over time.

Other techniques of local examination are broader in scope but similarly preserve the field of work. Simulation and trial

and error probe possibility through controlled variation, revealing how a configuration behaves under change. Induction and proof by contradiction challenge or extend results beyond their immediate context, while still maintaining the original field.

Higher-order families of techniques regulate orientation, continuation, and standards of success within an inquiry, and may be applied as needed to align ongoing work with purpose.

Signs, Signals, and Beacons

Signs

Signs are marks that can be interpreted, combined, or transformed. A sign can mark a boundary, note a change, record the presence of a feature, or signal a completed step. Signs indicate, without requiring additional explanation or interpretation.

Simple signs appear everywhere: a tick in a margin, a dot on a diagram, a label placed near a part of a figure, or a mark showing a measurement.

Signs prepare the ground for more elaborate forms of representation. They provide the minimal scaffolding needed to identify positions, states, or transitions.

TYPES OF SIGNS

Signs take different forms according to the relation they bear to what they indicate. Indices include traces, measurements, and simple markers. Indices point by direct connection or correlation. Icons, such as diagrams, sketches, and geometric figures, resemble the appearance of what they represent. Symbols in the strict sense discussed below are conventional marks detached from resemblance: their meaning is governed by rule rather than likeness.

SIGN SYSTEMS

In more formal settings, a sign system specifies allowable marks, as well as rules to form, use, and transform them. The Roman Numeral system itself is a sign system.

Signals

A signal is a singularity within the field of signs. Anomalies, inflection points, gradients, fluctuations, resonance spikes, and failures of alignment all function as signals. They indicate where a procedure is emerging, changing, or breaking.

Signs indicate direction. Signals indicate states. Beacons provide orientation.

Beacons

Beacons are high-level qualitative regulators. When encountered, as effects, feelings, or intimations, they command a change in the practitioner's strategic approach to the task. They fall into three conceptual divisions: formation, continuation, and validation.

FORMATION

Novelty and uniqueness are beacons of formation. They attract attention and signal the need for further examination.

CONTINUATION

Plausibility is a beacon of continuation that acts by intimating next steps. Obviousness is the limit of plausibility.

Equality establishes that two apparently different elements or relations are essentially the same. This identity allows the practitioner to substitute one for the other, leading to process simplification.

Persistence shows the recurrence or sustained presence of a set of relations throughout more than one step of the process. This recurrence provides a reliable basis for treating those relations as foundational and resistant to subsequent adjustment, or for predicting future behavior.

Intensity signals the energetic state of the engagement. A drop in intensity indicates a loss of contact with the object of study, signaling the need for re-engagement or a change of scale. Excessive intensity signals forcing, where the practitioner imposes relations rather than discovering them. It governs the vitality required to sustain the step.

VALIDATION

Beacons of validation include affinity, resonance, discord, obviousness, alignment, and gestalt.

Affinity signals kinship or fit among objects or relations, indicating that elements belong together and may be combined without distortion.

Resonance marks a moment in which relations reinforce one another, producing a path of interpretation within the inquiry.

Discord functions as a negative beacon, indicating incompatibility or unresolved tension that calls for revision of steps, scope, or framing.

Obviousness appears when a configuration presents itself as inevitable, suggesting that the current orientation is appropriate to its purpose.

Alignment indicates whether relevant elements have settled into a working configuration for further operations. Gestalt refers to the sudden appearance of coherence among previously separate parts, instructing decisive continuation of the work.

Equilibrium signifies the moment where conflicting or disjointed relations are aligned with purpose, confirming their

coherence.

Beacons provide diagnostic data to evaluate and adjust strategic movement through the course of inquiry. They have a distinct role in showing where re-composition is necessary, where continuation is possible, or where a compositional point of rest has been reached.

Signals are local atmospheric effects; beacons are global attractors.

Symbols

Rather than simply marking or pointing, a symbol can stand in the place of a relation or a step in a procedure. Once introduced, it can be combined, compared, or transformed. A familiar example is the letter x in an equation: it stands for a quantity, and operations can be carried out on x without knowing the quantity itself.

Symbols are the base layer of more elaborate representational systems such as notation, diagrams, and models. They make relations portable, repeatable, and open to transformation.

Exercise

An exercise is a trial execution, to see how steps work in practice. Exercise can expose dependencies, sensitivities, and points of failure.

Exercises can also occur within imagination, by rehearsal in thought.

Attunement

Attunement is a procedure of tuning or adjustment. It operates when repetition and variation reveal misalignment, imbalance,

or drift, allowing fine corrections. Attunement functions as a local regulator to maintain fit and proportion.

Aptitudes

Aptitudes are subjective background capacities that condition application. They influence intensity, pacing, sensitivity, and responsiveness, especially where judgment and attention cannot be automated.

Rigor maintains care for consistency and proportion as the work develops. Plasticity allows shifts of scale or shape when the current orientation no longer fits. Discrimination identifies differences for continuation of steps.

Repetition

Repeating steps deliberately can reveal persistent patterns, which are a signal of coherent progression. Recurrence demonstrates local reproducibility.

In performative methods, repetition is the mechanism by which skill, accuracy, and proportion improve.

In adaptive or self-correcting procedures, repetition provides the feedback that informs revision. The procedure learns from its own history because iterative trials make the relevant differences visible.

Variation and Rhythm

Variation is the complement of repetition. By changing one feature while keeping others fixed, a practitioner can observe sensitivity to conditions. Small variations show which parts of a procedure are essential; larger variations expose limits.

Rhythm is a practical balance between repetition and variation. Too much repetition produces rigidity. Too much variation produces noise.

Correlation and Alignment

When different observations or results share a recognizable pattern, they are in correlation. A sequence of measurements that rises and falls in the same way, or separate cases that display similar boundaries or transitions, show correlation. It acts as a signal.

Alignment is the act of placing elements in a comparable position. Aligning measurements on a common scale, arranging observations in a consistent order, or redrawing diagrams so that equivalent parts occupy similar positions are all forms of alignment. When the elements aligned are steps of a configuration, alignment becomes a measure of fit and orientation among those steps. Alignment can reveal correlation, among other patterns.

Qualitative Assessment

Qualitative assessment concerns whether a method is well-fitted to its purpose. It may be internally coherent yet poorly suited to its aim, or it may be well-suited yet unable to sustain continuation.

A method is fit for purpose when its application produces the results intended by the practitioner. There are degrees of fitness. A bad method simply does not do what it is intended to do, or worse. A trivial or useless method is one for doing something that is easier to do without it; nothing is gained by following its steps.

A method may be elaborate and still well-fitted if it serves purpose better than a simpler approach. It may be simple and

still adequate if the task calls for no more than a few operations.

Fitness for purpose is the condition under which a method has utility. A method that is fit for its purpose yields some gain relative to not using it at all; a method that yields no such gain is trivial or useless. Utility is the practical expression of fitness for purpose.

Reproducibility

A method is reproducible when others can apply it, and it yields results that remain comparable. Reproducibility usually involves variation in conditions rather than changes in the method itself; yet under changing conditions, revision may be required in order to continue producing comparable results.

A musical score specifies relations rather than outcomes. It fixes intervals, durations, proportions, and cues, but it cannot fix the conditions under which those relations will be executed. Those conditions necessarily vary: instruments differ in range and timbre, acoustics shift, and there may be changes in tuning and tempo. To preserve comparability of results, the manner of execution must sometimes be revised. The score remains reproducible by sustaining relations under variation.

Atlas of Methods

As used in this Register, the Atlas is a surveying device. It identifies a small set of features along which differences in methods appear in predictable ways. These include their aim, the kind of evidence they depend on, and the type of reasoning they use. Further distinctions concern how results are expressed, how reliability is established, and the scale at which the procedure operates. These features serve as dimensions of description.

By setting them out along shared dimensions, the Atlas can reveal shared features that are not apparent when considered

in isolation, in the same manner that systematic arrangement in a Linnaean table exposes affinities and distinctions among specimens.

This device also assists construction. When methods are revised, combined, or adapted, the Atlas provides a way to record the result and to compare it with existing configurations. It is a reference against which new configurations can be set.

The description of the Atlas is developed in further detail in Appendix A.

Summary of the Grammar

The grammar establishes the elementary gestures through which inquiry begins, the operational mechanisms that organize relations, and the semiotic devices that orient analysis.

PART III — OPERATIVE MODELS AND TRANSPOSITIONS

An ideational ‘machine’ as described in the Register is a constructed arrangement of relations that, once set in motion, produces intelligible consequences according to its internal organization. It takes inputs into a model and yields outputs that were not individually specified in advance.

Toy Models

“The human being striving for rationality and restricted within the limits of his knowledge has developed some working procedures that partially overcome this difficulty. These procedures consist in assuming that he can isolate from the rest of the world a closed system containing only a limited number of variables and a limited range of consequences.” (Herbert A. Simon, *Administrative Behavior*, 4th ed., 1997, 94)

Toy models are simplified constructions used for examination in a reduced setting. Physics, mathematics, and the humanities all rely on such constructions to isolate and focus on key relations that may be more difficult to discern in full context.

Cavendish’s Spheres as a Toy Model

Perhaps the most well-known example of a toy model is Cavendish’s torsion experiment. Cavendish’s model isolates a small domain whose elements and relations can be examined without distraction. Two small spheres are suspended from a rod; two larger spheres are placed near them. The positions

of the spheres define the elements of the domain, and the measurable differences among these positions define the relations. When the larger spheres are moved, the arrangement changes in a controlled way, and the response of the suspended rod tracks the effect of that change.

The model restricts itself to a narrow set of variables: the distances between spheres, the orientation of the rod, and the measurable deflection that follows from the new arrangement. Each movement alters the orientation of the apparatus and brings new relations into view. Veracity is determined through the fit between the observed deflection and the change expected from the new arrangement. Drift is limited by the controlled environment; correction is possible because the relations are persistent and repeatable.

This reduced setting exposes a principle: small positional changes in one part of the domain can induce a measurable effect elsewhere. Cavendish constructed the arrangement so that the relevant differences were clean, sparse, and recoverable. Cavendish's spheres serve as an exemplary toy model: a controlled reduction that reveals how a change in relation yields a change in orientation, all within a domain simple enough for each step to be followed with clarity.

A toy model is a machine for thinking: simple enough to manipulate, strict enough to instruct, and rich enough to meaningfully assist the transition from intuition to method.

Representational Transformations

Some operations in method reorganize a field by altering the form in which its relations are presented. These operations are representational transformations. They provide alternate views that make certain relations explicit or expose dependencies that are difficult to notice in the full field.

In algebraic diagrammatic rewriting, an expression written as a line of symbols is redrawn as a diagram. Terms appear as nodes, operations as connections, and repeated sub-expressions as shared structures. The mathematics remain the same but the relations are presented visually. Patterns that were buried in the line of symbols may now be obvious at a glance.

Projections

A projection selects a relation-pattern in a complex field and represents it as a reduction that exposes its dependencies. A toy model is a projection. It strips away detail while preserving selected relations, making a dependency, tension, or symmetry easier to grasp.

In principal component analysis, a high-dimensional dataset is projected onto a lower-dimensional subspace so that the dominant directions of variation become visible. The projection removes most coordinates, but it preserves the dependency that orders the variation. The lower-dimensional space is incomplete, yet it reveals one relation, shared variance, that would be difficult to discern in the full field.

Worktables

The agent or practitioner constructs a worktable by turning attention to a domain and designating an object, a group of objects or relations, or a region of the domain for study. The agent may treat the domain itself as an object. This designation already alters the field: the observer transposes material from its original setting into a region set aside for study.

The agent chooses what to bring forward, how to present it, and what to leave aside.

The worktable changes as the inquiry proceeds. After observing, inspecting, or analyzing the worktable contents, the

practitioner may return to the domain to select additional objects or pre-existing arrangements. The agent may add these to the worktable and examine them separately, or may place them aside for later use. The worktable can be reorganized at any point. For this purpose, the items are sorted by lists, placed in small tables, set into arrays, or displayed in a simple matrix. These are containers that hold the material in a fixed arrangement for comparative analysis.

As the work proceeds, the agent may need to develop higher order containers to organize multiple configurations. These are superstructures. Examples in the Register include the *n*-tuple matrix and the MusicBox: each imposes a deterministic configuration of multiple objects the manipulation of which reveals relations not apparent when the material lies unarranged.

A worktable must fit the present phase of the inquiry. A minimal worktable may hold only a few distinctions. A more developed one may contain several related arrangements. When the table becomes cluttered, or no longer matches the direction of purpose, the agent must clear or rebuild it.

Containers

A container is a simple construction that gathers material for inspection. It holds objects in an arrangement long enough for comparison, rearrangement, or further division. A note, a list, a table, a small array, or a marked diagram each serve this purpose: they provide a place for initial arrangement.

A container also anchors the beginning of an inquiry. A few rapid notes can capture the first distinctions before they scatter. A small matrix can crystallize the early pattern of a problem and allow later steps to be taken from a fixed point. When the work returns to these early markings, the observer can see how the field has shifted: new relations appear, old ones fall away,

and some regions become clearer.

Containers grow or contract with the movement of the inquiry. A list may be reorganized as themes converge. A table may be expanded to include new cases. A small matrix may be rotated or repartitioned to reveal a different alignment. Each revision of the container reflects a new orientation and prepares the next set of steps.

Alongside notes, sketches, tables, arrays, and simple matrices, several other devices serve as early containers within method. A short sequence written in a margin can hold the order of events while the relations among them are still unclear. A small diagram, such as a triangle, a circle, or a rough lattice, can steady the first sense of proportion or branching. A column of examples gathered from different contexts can expose a family resemblance that is not yet named. A brief schema of a few positions arranged on a page can hold competing distinctions until testing of their alignment. A timeline drawn with only a few marks can preserve the order of steps long enough to note inflection, delay, or return.

These small constructions give the inquiry an early surface on which differences stand beside one another for examination.

Superstructures

A superstructure is a construction that fixes relations in a configuration so they can be examined, compared, and revised without dispersing. It differs from a container. A container holds material; a superstructure imposes an arrangement. The n-tuple matrix is the first superstructure in this Register that reaches beyond simple arrangement and introduces a formal internal logic. It allows several dimensions of a method to be examined together.

Matrices lend themselves naturally to the organization and

study of arrangements, through juxtaposition, rotation, and transformation of elements. An n-tuple matrix begins with a familiar idea: a table whose rows and columns index the objects or relations under study. But each entry is not a single value. It is an n-tuple, a fixed grouping of related elements drawn from the material under study. One coordinate of the n-tuple may record conceptual relations. Another may record performative steps. Others may record contextual conditions, or any other dimension relevant to the inquiry.

The MusicBox follows the same sequence. It is a specialized n-tuple matrix with its own choice of coordinates and rules of combination.

The MusicBox

The practitioner begins by selecting one or more musical scales and expressing each as an n-tuple of its pitch intervals within a matrix. Each scale is assigned a column position, and each row lists a cyclic permutation of the interval sequence.

By placing the interval sequences in these fixed positions, their patterns stand in visible and acoustic relation to one another. The practitioner can read each rotation against the others.

The MusicBox extends cleanly by assignment of harmonic frequencies to pitch degrees. Each interval in the n-tuple is assigned a corresponding harmonic coordinate. When a permutation is applied, the harmonic coordinates follow the same re-indexing as the intervals. This construction brings harmonic profiles and interval relationships into the same field, and a simple acoustic device can trigger the frequencies in any selected entry to make the profile audible. The extension adds a further coordinate but leaves the underlying configuration intact.

Resonance occurs when two n-tuples share intervals and fit together. A shared core or a reinforcing sequence may produce this fit. A rotated mode in one scale may meet a rotated mode in another. When the patterns do not fit, discord stands out clearly.

Attunement guides the practitioner in the choice of the next entry to read or play. The practitioner follows a sequence drawn from the scale at hand: for example, by selecting the next rotation that preserves the same interval set, or another scale that shares specific intervals. When two adjacent entries share the same intervals or repeat part of the same interval sequence, the visual and audible relation is evident. When the next entry introduces a different set of intervals, the change is immediately perceptible.

The MusicBox provides an organized field in which several scales and their rotations can stand together long enough in a variety of selected arrangements for resonance, attunement, and systematic exploration to become directly readable.

Plausibility Machines

A plausibility machine is a constructed system used in methodical work to engage a tentative configuration.

Its role is to place a configuration into operation so that the practitioner can judge whether further steps can proceed in alignment with present purpose and intent. A plausibility machine is a device that tests whether a configuration is workable enough to warrant continuation, prior to proof or validation.

The machine shows whether the configuration holds together under repetition or motion and whether subsequent steps remain possible. By externalizing the configuration, the machine allows the practitioner to adjust orientation as the

work proceeds, responding to what the configuration makes evident. A successful outcome allows the practitioner to continue. The configuration is taken forward into further methodical development.

A piano itself is a plausibility machine of harmony, as is the MusicBox matrix discussed above. Each permits the practitioner to hear at once whether a harmonic relation works.

A whiteboard is a simple plausibility machine. It provides a surface on which tentative relations can be set down, rearranged, and extended. The practitioner places marks, symbols, or diagrams on the board to represent a provisional configuration. As additional elements are added, erased, or repositioned, the configuration is exposed to continuation in real time. Relations that can be carried forward remain legible under extension; relations that cannot sustain further steps become crowded, inconsistent, or lose orientation. The board makes these limits immediately visible. Because marks can be altered without cost, the practitioner can adjust the configuration repeatedly, testing whether further steps remain possible while maintaining orientation toward purpose.

A four-bar linkage mock-up used for kinematic testing functions as a deliberately tailored plausibility machine. This is a simple physical mock-up of a mechanical linkage, built from flat pieces of wood, cardboard, or plastic joined by pins or bolts. Each piece represents a rigid element, and each joint represents a point of rotation. The proportions are chosen to match a proposed relation among lengths and angles, not to meet production tolerances. The mock-up is constructed so that it can be moved by hand. No motor is attached. No load is applied.

The practitioner applies motion directly, guiding the linkage through its full range of movement. The purpose of the device

is to allow the proposed relation among parts to operate in the open, under the simplest possible conditions. When the linkage is moved, its behavior becomes immediately evident. In some cases, the motion proceeds smoothly through a complete cycle. The parts move in a regular way, and the relation between input motion and output motion is clear. In other cases, the linkage binds, locks, flips orientation unexpectedly, or collapses into an unintended configuration.

These outcomes are encountered directly in the hand. The device checks whether the proposed arrangement of parts can sustain motion at all. If so, the relation among the elements is workable enough to be carried forward into further development.

A plausibility machine provides a quick internal check: Is this pattern desirable and repeatable, or is it just noise?

Go and the Grammar of Method

A Go game begins with an empty board and proceeds by placing stones one at a time. Each stone alters the board configuration, which consists of relations of adjacency, connection, and enclosure.

A player selects a move with purpose and intent. From the standpoint of play, the significance of a move lies in how it alters the arrangement.

The opponent then plays a move, introducing a further change. This may disrupt existing relations and may establish new ones. The player selects a subsequent move based on the updated arrangement.

A move's adequacy is undetermined at the moment it is played. Adequacy becomes apparent through subsequent play.

In Go each move responds to an evolving configuration shaped by the actions of both players. Novelty results from the

sequence of adversarial moves.

Method is essentially identical. Work proceeds from a configuration, which is altered by each new step. Factors beyond the practitioner's control may introduce further changes. Novelty results from sequences of steps taken in response.

In Go and in method, the practitioner proceeds by selecting a next step conditioned by the relations that remain coherent after the alteration.

Go embodies method. The game makes visible, in a purified model, what method consists in: purposeful action exposed to uncontrollable eventuation, whose adequacy is determined through continuation.

Adaptive and Self-Modifying Methods

Adaptive methods alter their own procedure during execution. The change is based on information generated by the method itself. The revision is internal and rule-governed.

Certain scientific procedures illustrate this type of adaptation. In computing, adaptive filters adjust their internal weights while processing a signal, so that each pass modifies the next. In engineering, adaptive controllers alter their correction parameters as system behavior shifts, maintaining controlled behavior as conditions change. In physics, adaptive optics update the correction pattern applied to an image as new measurements of distortion arrive.

Self-reflexive methods are a narrower class within this broader group. In these, the activity carried out is also the activity being modified. In piano practice, repetition, listening, and variation occur together, and the method of playing alters itself through performance. The study of methods itself offers another: examination and construction coincide, and the act of inquiry revises the procedure by which the inquiry proceeds.

Adaptive methods change themselves through use, and their outcomes become part of the procedure that follows.

Generativity

Generativity is the capacity of a method to enlarge its own field of action through the normal use of its rules. A generative method produces differences or steps that were not present at the outset. The method does not change; the enlargement results from the accumulation of its own operations.

L-Systems

An L-system begins with a small set of symbols chosen by a modeler. Each symbol functions as a mark with a defined role in a construction. The modeler also prepares a set of rewriting rules in which each rule assigns a replacement sequence to a symbol. The modeler then arranges several of the symbols in a short sequence. This initial sequence is the starting word.

In each step, an automated procedure scans the current sequence and replaces each symbol that has a rule with its assigned replacement sequence, leaving the others unchanged. This produces a new sequence, usually longer than the previous one. Each new sequence undergoes the same rewriting step; the procedure remains fixed throughout.

In a botanical drawing program, the symbols would have simple geometric meanings. 'F' may instruct the drawing program to move forward and draw a short segment; '+' may instruct it to turn left by a fixed angle; and '-' to turn right by that angle. When the rewriting rule replaces 'F' by 'F+F-F', the resulting sequence simply becomes a longer list of these drawing commands. After several steps, the process yields a long sequence containing these drawing symbols, which expands into a detailed instruction for a complex plant-like figure. The

apparent growth results entirely from the repeated rewriting of the sequence and the simple geometric assignments given to its symbols.

A method exhibits generativity when repeated use of its procedures yields an enlargement of its arrangements.

The Modular Gesture

Some methods work by assembling their results from small, repeatable components. A basic unit is identified, and a rule determines how that unit may be joined to others. When the same rule is applied, the components cohere. The method relies on the disciplined repetition of an element.

In a tessellation, one tile produces a complete surface when combined by a fixed adjacency rule. When a shape is placed according to the rule, each placement limits what can follow. Some shapes permit the process to continue indefinitely. A square can be extended throughout a surface without gaps. A regular pentagon cannot. A shape that is only slightly misaligned may appear to work at first, but the misalignment accumulates and eventually prevents further placement.

The method is tested by whether it can be carried forward under its own rule. The outcome is determined through continuation.

The Mandelbrot set shows the same principle in an iterative model. A single rule,

$$z \mapsto z^2 + c,$$

applied repeatedly, produces a field of behavior from minimal data. Each iteration uses the same operation, and the overall configuration results from the accumulation of these local steps. The fine detail of the boundary at every scale is the result of

repeating one transformation under fixed conditions. The complexity of the figure is produced by the repeated application of the rule.

As a method, modularity shows how inquiry can proceed by using a well-defined component and a consistent rule that governs how the work extends. The component and the rule together create a setting in which larger models can be built without losing clarity.

Structural Transpositions

A structural transposition shifts relations under inquiry into another domain where they make available a wider range of inferences and constructions.

Transpositions differ in the extent to which relocated relations can be reused for further steps and in what they preserve and what they discard.

Types of Transposition

The simple act of measuring transposes relations drawn from experience in a repeatable way. Stretching a rope, opening a hand, or pacing a distance establishes a reference that can be repeated and shared. Arithmetic also performs a transposition. Through counting, quantity is fixed in symbols that permit further steps.

Once relations are fixed symbolically, they can be relocated again into more formal settings.

In an isomorphic transposition, relocated relations are preserved in all functional respects. Measuring and counting already have this character: relations fixed by these acts can be compared, combined, and reused without alteration. In linear algebra, which is itself a structural transposition from arithmetic, a polynomial can be associated with a matrix. In

the companion matrix, roots correspond to eigenvalues, and factorization to invariant subspaces. Solutions to polynomial equations can then be derived seamlessly from matrix operations.

In a homomorphic transposition, relocation preserves relations among a limited range of steps. In music theory, a harmonic progression is usually first composed as pitches and chords in a particular key. At this stage, pitch height and range are fixed. The progression can then be rewritten using numbers or functional labels. In this representation, absolute pitch is removed. With pitch removed, the progression can be extended, continued, or modified without reference to any particular key.

Numerical harmony assigns the same symbol to any chords whose roots lie at the same interval from a chosen tonic, thereby discarding absolute pitch. Performance specifics such as voicing and instrument range are also omitted, leaving only interval-based progressions. In performance, the numbered progression is written back out as pitches in a specific key. When this is done, questions of range and physical feasibility reappear and can be checked directly.

In some transpositions, relations are relocated as patterns rather than as operations. The positive Grassmannian and its representation by ‘plabic’ (planar and bicolored) graphs discussed below provide a recent geometric example. A combinatorial representation allows manipulation and comparison of graph configurations that correspond to geometric relations, even though the original operations on those objects are not preserved.

In the simpler setting of musical sound production in performance, a musical score preserves relational sequences of pitch intervals, rhythm, and harmony, but the score itself is silent.

Metaphors are a way relations can be transposed without formal procedures. A metaphor allows people to notice similarities, anticipate behavior, or coordinate action by relying on shared cues. Those cues can guide thought, conversation, and judgment.

When we talk about a conversation ‘stalling’, nothing mechanical is happening. No engine, no process, no formal rule has failed. What carries over is a pattern: motion slowing, loss of momentum, difficulty restarting without intervention. These cues let participants judge when to change topic, introduce a question, pause, or stop entirely. The metaphor indicates an orientation that affects what someone might do next.

Reflective transposition allows translation in both directions, so that results obtained in one domain can be carried back to the other. The Riemann–Hilbert correspondence explained below exemplifies this reciprocity: analytic differential equations and topological descriptions by perverse sheaves can be translated into one another, permitting methods from each side to be applied between domains. Each description informs and reorganizes the other.

Transposition can act on objects, relations, procedures, symbolic regimes, or entire methods. It is a technique of inquiry when essential relations are preserved and the new setting discloses arrangements hidden in the original.

Case Studies

The following case studies draw on mathematics and physics, where transposition has long been made explicit and formal, and where problems are often approached by deliberate changes of representation. Each case concerns a landmark problem whose resolution reshaped its field and depended on a change of representational domain.

Riemann–Hilbert correspondence

The Riemann–Hilbert correspondence provides a canonical example of a reversible transposition, relating systems of linear differential equations to topological descriptions in terms of sheaves. On the analytic side, one studies differential equations together with their singular behavior and the local spaces of solutions they admit. On the topological side, the same data is encoded in sheaf-theoretic objects defined on stratified spaces, which record how locally defined information is continued, related, and assembled between regions and singularities.

The passage from differential equations to sheaves is a sequence of formal replacements. A differential equation is first replaced by its local solution spaces on small neighborhoods, and these local spaces are then organized together with rules governing analytic continuation and compatibility on overlaps. The resulting sheaf captures the global behavior of solutions without reference to the differential equation itself.

This correspondence allows analytic problems to be reformulated in topological terms, altering both the kinds of questions that can be posed and the tools available for addressing them. Conversely, topological descriptions can be interpreted back in terms of differential equations, their solutions, and their singular behavior. The Riemann–Hilbert correspondence demonstrates how work carried out in either description can be translated, recovered, and extended in the other.

Positive Grassmannian

Lauren Williams and collaborators have shown that the positive Grassmannian, a geometric space defined by positivity conditions, can be described using discrete combinatorial objects. In this approach, regions of the geometric space are

associated with planar graphs, permutations, and related combinatorial descriptions. The geometry appears as a collection of regions with well-defined boundaries, while the combinatorics represents these regions through graphs and permitted transformations between them.

The geometric description can be translated into a combinatorial one by replacing continuous regions with discrete patterns and geometric relations with descriptions among graphs. Although this translation omits much of the analytic detail of the original space, it preserves the dependencies needed to study total positivity and scattering amplitudes. This provides an example of homomorphic transposition: a change of representation that preserves what is relevant for interpretation and computation.

Wiles and modularity

Andrew Wiles' proof of Fermat's Last Theorem passes through a chain of transpositions. A statement about integral solutions to a Diophantine equation is recast in terms of elliptic curves, then related to modular forms, and finally expressed in terms of Galois representations. The original problem is shifted from arithmetic into analytic and geometric settings where relationships not visible in its original formulation can be examined directly.

Each link in the sequence of transpositions carries forward enough of the objects under study and the relations between them to allow the next reformulation to be carried out without loss. The Diophantine equation becomes part of a larger modularity conjecture. The conjecture becomes a statement that two independently defined descriptions of the same object agree. The proof establishes the modularity, and the original number-theoretic statement follows as a consequence.

These examples show different degrees of transposition, ranging from cases where expressions can be converted back and forth without loss, through cases where distinct objects are deliberately identified, to cases involving several successive reformulations.

In each case, the method proceeds by recasting the problem into a space in which the relevant relations can be explicitly identified and manipulated.

Formal Representation of Transpositions

Some structural transpositions preserve enough operational identity between domains that they can be treated as mappings between formally represented methods and, in more developed cases, systems of methods (Part V).

When a transposition carries relations and procedures of application in a way that remains comparable under restriction and extension of domain, it may be represented as a functor between method-assignments in the sense of Annex I and Annex II.

In such cases, the relocated methods can be examined through sheaf-theoretic, algebraic, and geometric constructions, allowing formal analysis of how applications compare and diverge as domains change.

PART IV — MACHINES

Fixing Instruments

Fixing instruments fix relations in place so that they can be examined, transformed, and reused.

Some of the earliest examples are simple numerical instruments: tally marks, counting rods, abacuses, bead frames, and early calculating boards. Slide rules and nomograms measure relations among quantities in fixed geometric form, allowing proportional or functional relations to be read directly. Paper tables of logarithms or trigonometric values preserve complex calculations in a durable surface that can be scanned and applied without recomputation. Coordinate paper, grids, and mapping frames provide a field in which spatial or quantitative relations can be located and compared. Instruments such as microscopes and telescopes stabilize relations of scale and visibility, holding configurations steady for inspection. Looms and printing presses give durable form to patterns or sequences, allowing them to be reproduced, inspected, or transferred.

Orientation Instruments

A straightedge and a geometric compass, together, are a canonical example of instruments that both fix and orient.

Orientation instruments supply bearing, scale, and direction, allowing a domain to be entered and traversed without losing the relation among its parts. Their role is to mark position, trace paths, and make movement intelligible.

A magnetic compass sets direction against the earth's magnetic field, giving a reliable reference for movement through a landscape. The astrolabe links celestial positions to terrestrial angles, allowing latitude or time to be inferred from the height

of a star. The sextant refines this capacity by measuring angular distances with sufficient precision to determine position at sea when combined with a reliable timekeeper. The chronometer gives the reference needed to convert local observation into a global coordinate; and tables of lunar distances, star charts, and almanacs extend these instruments by supplying the background relations they presuppose. Surveying chains, sighting tubes, and other simple tools anchor local orientation by giving fixed points from which distances, alignments, and elevations can be established.

Each of these instruments carries a spatial or temporal relation into a physical medium that can be consulted repeatedly.

Grand Machines

Grand machines are devices that operate through complex arrangements of relations, sustaining method at scales otherwise inaccessible directly.

The printing press is an early example. It fixes a sequence of symbols in a repeatable pattern and transfers that pattern to the page with uniform pressure. The machine gives text a durable and reproducible form, allowing ideas to be carried far beyond the place in which they are composed. The loom extends this capacity into material patterns. It organizes threads into an ordered arrangement that can be repeated with precision, making patterns visible in the fabric itself.

Optical and exploratory machines carry this development further. The microscope extends the scale of perception by controlling the path of light through a fixed arrangement of lenses, revealing forms that cannot be seen with the unaided eye. The telescope does the same at a distance, gathering faint light into an image that can be inspected and compared. Satellites provide a platform in which position, orientation, and

measurement can be maintained outside the earth's surface, using the surrounding space as a field of observation. These machines give access to relations that would otherwise remain out of reach.

Scientific instruments of increasing size and precision reveal relations by staging controlled interactions and recording their outcomes. Cyclotrons and colliders accelerate particles along defined paths and bring them into collision. The resulting traces guide the determination of relations governing particle behavior. The computer extends this principle into a logical domain. It arranges switches and circuits so that the state of each element determines the next; sequences of such transformations become a basis for symbolic work.

Wendelstein 7-X, a stellarator built to test whether a plasma can be confined by a geometrically determined magnetic field, carries this progression to a higher scale. The shape of its magnetic coils is determined by calculations that solve the equations of motion in advance: the machine is at once physical and mathematical. Every curve of the device reflects formal limitations dictated by topology and differential geometry, but the resulting configuration tests whether the geometric conditions required in theory for plasma confinement hold when realized in physical matter. In Wendelstein 7-X, the physical artifact becomes an enactment of a geometric idea.

Grand machines make it possible to carry out operations that exceed the capacity of direct thought.

Ideational Machines

An ideational machine, introduced previously as an operative model with inputs and outputs, is considered here in terms of what makes that operation possible. It is a construction that fixes relations in advance of any particular application, so

that informal or heterogeneous material is converted into an organized field in which that material can be examined under explicit rules. It imposes an order on what can be built and compared. Its function is to organize the space of possible constructions so that movement within it remains coherent.

Fuzzy Causal Set

A Fuzzy Causal Set is an ideational machine that models causality as graded implication rather than binary determination. It consists of a set C of elements—such as events, propositions, perceptions, states, or conceptual units—together with a function

$$\mu : C \times C \rightarrow [0, 1],$$

where $\mu(x, y)$ expresses the degree to which x contributes to, or aligns causally with y .

Causality is treated as a field of weighted relations capable of expressing partial influence, overlapping causation, ambiguity, and unequal causal contribution.

The fuzzy causal set functions as an ideational machine because it converts qualitative judgments of relevance or influence into manipulable relations.

Virtual Spaces

Virtual spaces are examples of ideational machines that construct a setting in which new forms appear. They allow thought to move through theoretical spaces.

A clear example of a virtual space appears in phase space. In classical mechanics, motion is represented as trajectories in a domain whose coordinates are position and momentum. In statistical mechanics, ensembles are defined as distributions over the same domain.

No such space is ever observed. It is constructed so that relations among quantities can be treated as relations among points and paths. Once constructed, changes in a system appear as movements, and stability, recurrence, and divergence become visible as geometric features.

Phase space is determined entirely by the relations that define the system. Its structure is not derived from perception but from the formal organization of variables. In this virtual space, abstract relations among quantities are arranged as visible positions and paths.

A spherical mirror is introduced here as another virtual space. It behaves as a manifold of reflection and distortion defined by treating a mirror surface as having a spherical curvature. The points and geodesics of the surface result from the rules that describe how an image is reflected within it. The reflection space may be characterized as a virtual Riemannian manifold, within which relations can be examined as cast inside a curved surface; yet the surface is a formal construction. A fuller formal account of this construction is developed in Annex III.

Virtual spaces provide domains in which method can proceed without being tied to physical space. The relations involved define a field of possible, but constructed, states in which methodical steps can be taken.

PART V — SYSTEMS OF METHODS AND METHOD SPACE

Systems

A system of methods combines methods themselves into larger arrangements. In such systems, methods operate in various degrees of coordination. Recognizable arrangements can be revealed when methods are repeatedly combined in the same order. Branching occurs when one or more methods continue the work in different directions. Additional structure develops when the results of the methods are combined into new configurations. Relations established within one method can be carried forward, transformed, or extended through others.

Formal axiomatic systems in mathematics are deliberately constructed arrangements in which definitions, axioms, and rules of inference are bound together. Proofs allow later arguments to rely on earlier ones as points of departure. Because relations established in one domain can be carried intact into another, these systems transport inference through varying contexts.

Common law judicial systems operate through accumulated decisions that remain active within later judgments. Individual rulings enter into a network of precedents that shape how future cases are interpreted and resolved. Adjustments occur through appeal and reinterpretation, allowing the system to respond coherently to new circumstances.

In formal systems, coherence depends on strict agreement among local applications. In less constrained, adaptive, or exploratory systems, coherence may be maintained without such agreement, through reorientation and adaptation.

System Coherence

The combination and extension of methods in a system increases the scale of inquiry. The maintenance of coherence in the system is indicated by several signs. Alignment distributes itself throughout the system rather than remaining local. Relations established at one level resonate with those at others. Certain patterns persist as the work shifts scale or complexity, and the system can be refined or adjusted without collapsing.

A coherent system is a unity capable of operating at different scales without fracture.

Interoperability

Two systems interoperate when results produced in one can be transposed into another, and when patterns of inference can be imported or exported without distortion. Interoperability requires that transpositions preserve coherence and that alignment can be maintained as the work continues in different scales or domains. It is essential for interdisciplinary inquiry, for unifications within mathematics, and for scientific modeling.

Failure of Systems

Systems fail in several characteristic ways. A system may become rigid when it can no longer be adjusted for changes in its field. It may be sealed against feedback and lose the capacity for correction. It may fail to generate useful results when applied where it does not belong, and it may drift when it is no longer oriented towards its purpose.

Failure at the system level is more consequential than failure within a single method because the breakdown propagates throughout the sub-parts that depend on it.

Systems and Meta-Reflection

A system exhibits meta-reflective capacity when it can register tensions within itself and adjust its internal relations in response. It may adjust its orientation when its direction no longer matches its purpose, and it may revise the interfaces through which it exchanges information with other systems.

An autonomous robotic cleaning system operates through a coordinated set of methods: surface detection, debris identification, and cleaning execution. As the environment changes, the system registers deviations through its sensors, and automatically triggers adjustments to path planning and task sequencing.

A packet-routing network similarly operates through a coordinated set of methods: congestion detection, cost estimation, and update propagation. As network conditions change, deviations are registered internally through routing metrics, triggering revisions to allow redistribution of traffic without external intervention.

Method Space

Method space is the domain in which method becomes its own object. The study of methods as objects within a defined space focuses on their structures, transformations, and expansions of domain. It concerns systems of methods, relations among methods, and operations upon methods.

A more detailed description of method space is given in the Atlas of Methods (Appendix A).

Arrangements of Methods

Situating methods within a domain makes it possible for them to be arranged as objects themselves. Relations among them

become visible, and the methods can be differentiated and compared. This is the essence of methodical study, but applied to methods themselves.

Meta-Structures

When methods are placed in relation to one another rather than treated in isolation, larger patterns become visible. Related approaches may cluster into families that share a common grammar or representational orientation. In more complex settings, several approaches interact as ecosystems, supporting one another over time. Some form lineages, developing through generalization or transposition between domains. Others take the form of machines, in which execution is partly automated and modification becomes possible. Taken together, these arrangements define fields of possible methods, organized by parameters such as symbolic regime and range of transposition.

Evaluation

Evaluation of methods as objects themselves concerns how they behave under use and change, rather than whether their outputs are correct in isolation. A method may be judged by whether it gives rise to further methods or lines of inquiry, or by whether it remains intelligible when transposed into another representational regime. These qualities are exhibited in the operative models discussed in Part III.

Formal Evaluation and Testing

Methods admit formal representation. Annex 0 shows how even early, informal, or “one off” methods can be represented algebraically.

Many methods can be applied in more than one setting in the same way, so that different uses of a method can be set side by side and examined together. In more complex settings, methods are combined into systems whose components persist and interact between domains. More advanced formal modeling applies to such methods and systems whose operations remain stable under variation of input and under restriction or extension of domain, and whose different applications can be compared as instances of a continuing procedure.

These properties make it possible to model such methods and systems through more focused algebraic, geometric, and sheaf-theoretic constructions. These constructions make visible relations and failure modes that the minimal algebraic representation leaves indeterminate.

A method suitable for advanced formal representation persists through variation of context. A system suitable for such representation maintains recognizable patterns of interaction among its components as conditions change. This persistence allows behavior to be followed throughout refinement and use.

The annexes of the Register develop formal representations for distinct regimes of method behavior. Annex I describes the relation between local applications of methods and their behavior over composed domains. Annex II concerns the algebraic composition of methods and sub-methods within systems. Annex III models alignment and orientation where these can be expressed geometrically. Each annex corresponds to a specific mode of formal description.

Just as methods can be compared informally as arranged objects within a space, formal evaluation operates by arranging applications and interactions of methods into comparable relations.

Appendix B introduces a testing discipline for such methods

and systems. It specifies how formally represented methods may be evaluated through local alignment, veracity, and uncertainty conditions, composite tests, and a global alignment functional, and how continuation and failure may be assessed.

Appendix C applies the constructions developed in the annexes to a specific case study, demonstrating the diagnosis of compositional incompatibility in a mechanical system.

The formal models described in Annexes I–III are well suited to cases in which applications and components of a system can be related through equivalence classes (Annex 0), structural transpositions (Part III), and persistent patterns of interaction. When such relations are available, the annexes make it possible to examine systematically how methods compare as domains and conditions vary.

Through the arrangement of methods and their applications as objects in a domain, they are brought back under the foundational axiom that meaning arises from the arrangement of relations.

In the intentional accumulation of steps of methods and the arrangement of methods themselves, meaning is both constructed and discovered: it is configured.

PART VI — METAPHOR

In methodical terms, a metaphor is a container for an operation of transposition in which a word, a phrase, or an arrangement of words and phrases is used outside its literal domain and applied elsewhere.

Words are relations within the domain of language: patterns of use, contrast, and implication. Meaning arises from the arrangement of those relations.

When a word or phrase is transposed, what moves is a relation, carried in linguistic form. A simple metaphor transposes a word or phrase. A play transposes many words and phrases arranged into scenes and sequences. The operation is the same, only the extent of arrangement differs.

An idea may be described as ‘fragile’. Nothing breaks, yet the metaphor renders the state of the idea perceptible: the sense that it cannot withstand pressure, that it may collapse under scrutiny, that it must be handled with care. The metaphor transfers the logic of fragility from physical objects to thought and, in that transfer, brings the idea’s actual state into clearer view.

A ‘sharp’ idea ‘points’ in a different ‘direction’.

Metaphor and the Life of Method

Metaphor contributes to method by opening new regions of possibility through the re-description of a field in another idiom. It provides points of orientation from which relations become visible. It may also generate new relations.

Through the use of metaphors, a practitioner can find a first way to stand somewhere in a domain before its distinctions are known. This lets work begin without paralysis.

In many methods metaphors are embedded in execution and can govern timing, force, and posture. They are part of how a method lives and is transmitted and serve as a means of calibration by emphasizing intent or orientation.

This is true in many everyday crafts and sports, as well as in music, law, engineering, and other more formal disciplines. Most of these metaphors are so familiar that practitioners no longer register them as metaphors at all. They function as ordinary instructions. They describe with an innate imagistic logic when or how a step should be taken.

More formalized metaphors draw disparate elements into a conceptual group and may convey specific instructions.

Metaphoric language suggests ways for methods to proceed, even in the absence of explicit formal rules.

In the forensic analysis of a crash site or a digital intrusion, the practitioner establishes a 'beachhead' by identifying a single, immutable fact, a physical coordinate, or a timestamped log entry. This is a point of orientation from which other relations become visible. By securing this perimeter, those relations can now be calibrated by their distance from this initial foothold. A beachhead allows work to begin by providing a boundary that the field itself lacks. It is the first configuration that remains coherent under continuation.

In musical performance, the metaphor of the attack governs the timing and posture of the body before the action occurs. It re-describes the onset of a note as a physical strike. This is a means of calibration by emphasizing the intent of the onset; it demands a physical coiling of tension that a literal description of 'starting a note' does not. The metaphor conveys the intensity of the move, its velocity and force, ensuring the method is transmitted through a precise physical command by the player.

In martial arts, metaphors such as White Crane Spreads Wings transpose a mode of movement and force distribution from animalistic shapes into bodily execution. The relational pattern of the crane's anatomy specifically instructs the positioning of the practitioner's limbs, by imitation. The metaphor governs posture, tempo, and direction simultaneously. The practitioner inhabits the metaphor as the method itself. The metaphor preserves the arrangement of relations between the transposed shape and the practitioner's movement.

In each of these cases, a metaphor imports a logic of action from one domain into another.

Structured Metaphors

A structured metaphor preserves a relational pattern as it moves from one medium into another. The sending space is a methodic structure of relations, transitions, movements, and limits; the receiving space is a performative sequence of characters, actions, consequences, and events.

A play extends the operation of metaphorical transposition from language to action. Sequences of human conduct are lifted from lived social reality and arranged within the bounded context of the stage, where their relations and consequences can be examined as a whole.

Theater as a System of Methods

The re-enactment of scenes from human experience in a play is methodical. It proceeds by steps as an ordered sequence of actions and utterances specified by the script and realized in performance. Multiple participants act in relation to one another in coordination with the same sequence and timing, which persists in rehearsals, performances, and productions. A constitutive feature of a play is that the same arrangement

can be enacted again by different participants under different conditions without loss of identity. The play itself is a method.

Dramatic theater organizes the play through multiple methodical, non-reducible levels. It operates simultaneously at the level of script, performance, organization, and finance. Each of these levels has its own steps and its own modes of failure. Each governs a distinct range of action and possibility within the overall system. No degree of textual coherence compensates for failures of performance, coordination, or feasibility.

In its formal complexity, the dramatic system of methods stands alongside other highly developed methodical systems, such as legal systems, but it is also an early system. Greek tragedy is the first fully explicit instance of such a system, already mature when it appears in the record, which suggests a long prehistory.

Nonetheless, dramatic theater presupposes social capacities for coordination, role differentiation, persistence, and the maintenance of extended sequences of action.

At the beginning of Part I, method is situated in relation to human society itself: methods arise when humans collectively distinguish, arrange, and persist in the pursuit of common goals.

Dramatic theater echoes that origin while adding one further determination: humans are compelled to re-enact and stage their own patterns of behavior.

ANNEX 0 — AXIOMATICS AND ALGEBRA OF METHODICITY

Definitions

Observer

The observer is the one who conducts the inquiry. The observer supplies attention, selection, intent, and correction. Method becomes possible because the observer can notice differences, hold them long enough to compare them, and adjust the developing arrangement as required.

Domain

A domain is any expanse where things can exist, in the world or in thought.

Field

A field is the region of a domain the observer addresses in an inquiry.

Elements

Elements are things that exist in a domain.

Difference

Within a given domain, we say there is a difference in any of these cases:

1. between the domain with an element and the domain without that element,
2. between one element and another element,
3. between one collection of elements and another collection,

4. between an element (or collection of elements) and the rest of the domain.

Relation

A relation is any difference of the four kinds listed above.

1. Meaning and Relation

Meaning arises from the arrangement of relations. A relation is a difference, and an arrangement of relations is any set of those differences in a domain. When such an arrangement is designated for further study, it becomes the present configuration, and further steps in thought or action can proceed from it.

In this formalization, a configuration designated for continuation is termed a structure.

2. The Emergence Step ($0 \rightarrow 1$)

The first change that makes structure possible in any domain is the shift from no elements to one element. This shift creates the first difference: the domain-with-one-element differs from the domain-without it. Because difference is relation, this moment produces the first relation in the domain.

Before the emergence step can occur, the observer must already have directed attention toward a domain, establishing the field in which differences may appear. This act is epistemically prior to $0 \rightarrow 1$, though the first structural change occurs only with the appearance of a distinguishable element.

This emergence step ($0 \rightarrow 1$) is the earliest point at which anything can follow from anything, since there is now something that stands apart from what was there before. Later

structure rests on this initial difference.

3. Methodicity

When more than one element exists in a domain, the elements can assume different arrangements. Each arrangement presents its own set of relations. The observer determines which of these arrangements suggests a further step in method, according to purpose.

Methodicity begins when the observer selects relations from among the available arrangements that can continue, extend, or refine intent.

4. The Next-Best-Step Principle

The observer advances intent by taking steps. Each step makes new relations available for consideration.

The observer chooses the next step by selecting a relation they judge to be relevant to purpose. The emergence step ($0 \rightarrow 1$) is the earliest instance of this, and every further step follows the same pattern.

5. Necessary Conditions of Methodicity

Methodicity requires certain minimal conditions:

1. **Difference** — at least one difference from which a relation can arise.
2. **Relations** — more than one relation, so that arrangements can be compared.
3. **Purpose** — a direction of concern that lets the observer judge relevance.
4. **Intent** — engagement sufficient to carry purpose forward.

5. **Orientation** — whether the direction set by purpose has been maintained.

When any of these conditions fail, methodicity collapses.

6. *Formal Methodic Disciplines*

Arithmetic, geometry, logic, language, measurement, and navigation belong to a foundational cluster of methodicity. Each of these disciplines relies on purpose, intent, selected relations, orientation, and steps. Arithmetic and measurement proceed by taking steps in an ordered way; geometry advances by arranging relations in a deliberate sequence; logic tests selected relations against veracity; language names elements through selected relations, and navigation depends essentially on orientation.

These disciplines depend on method. Method can exist without them.

7. *Biological Methodicity*

Methodicity is not confined to human thought. It may appear wherever activity in a domain changes in relation to differences in a way that sustains coherence. A plant changes its growth in response to a difference in light. An animal shifts its movement when a nearby difference affects how it proceeds. A neuron changes how strongly its firing influences a connected cell when a change in its input alters how it is already acting: the connection becomes stronger or weaker the next time the neuron fires. Such changes meet the minimal form of a step: a difference appears, a relation shifts, and the next activity follows that shift rather than drifting from it.

8. Evolutive Methodicity

When a method is repeated, its steps do not remain identical. Small differences appear in the steps. Some of the differences maintain coherence; others do not. The ones that maintain coherence tend to recur, and the others fall away. As these differences recur, the method changes. Methods evolve.

9. General Form of Methodicity

Methodicity is the progression of transitions in a relational domain, initiated by the $0 \rightarrow 1$ emergence of the first usable difference, and sustained by next-best-step work that maintains or improves coherence. Methodicity appears in thought, skill, adaptive behavior, biological activity, cultural practice, computational work, and other domains shaped by relations.

The Algebra of Methodicity

The Algebra of Methodicity treats step sequences as elements of a free semigroup. Its quotients correspond to disciplines, where sequences are treated as equivalent when they produce the same configuration.

0. Orientation

Method begins with the appearance of the first usable difference. Once a difference is present, the observer can distinguish further differences, and steps become possible. The algebra below expresses that progression at its most economical, as the minimal formal pattern that any method must exhibit once distinctions accumulate and steps occur in sequence.

1. Steps as Generators

Let D be a domain. Let Δ be the set of differences in D that have been designated as usable. In the sense of Annex 0, these are relations that have become available as differences selected and retained by an observer for further steps.

A step is the act that compares differences or brings further differences into view. Once steps exist, they can follow one another.

Formally, let Σ be the set of steps available after the first retained difference. The free semigroup on Σ :

$$\Sigma^+ = \{\sigma_1\sigma_2 \cdots \sigma_n : n \geq 1, \sigma_i \in \Sigma\}$$

is the base algebra of methodicity, retaining only the concatenation of available steps.

It establishes that steps occur in sequence and that later steps proceed from the configuration formed by earlier ones.

This construction is minimal. It has no identity element and no inverses. It has only associativity, in that sequences can be extended. This is the algebraic shadow of method: nothing more is assumed.

2. The Effect of Sequences

A sequence of steps produces an arrangement in the domain. Define:

$$E : \Sigma^+ \rightarrow \mathcal{R}(D)$$

where $s \in \Sigma^+$ is a finite sequence of steps, and $E(s)$ is the resulting configuration.

Each step alters what can be seen, compared, or extended. Persistence and drift appear as empirical conditions:

$$E(s) = E(s') \quad (\text{persistence})$$

$$E(s) \neq E(s') \quad (\text{drift})$$

These are properties that result from repeated action in the domain.

3. *Disciplines as Quotients*

A configuration is the present arrangement of differences produced by a sequence of steps.

As inquiry develops, recurring sequences begin to cohere. Some are continued; others collapse into equivalence or cease to support further steps. A discipline is a domain in which step sequences are treated as equivalent when they produce the same configuration.

Let $\equiv$ be the equivalence relation generated by step sequences that lead to the same configuration. Then a discipline appears as the quotient:

$$\Sigma^+ / \equiv$$

This construction presupposes that the domain contains reasonably stable and comparable configurations, as in Go, chess, program states, or formal proofs.

The quotient selects for genuine differences in step progression by excluding variations that do not change the resulting configuration.

A discipline can be abstracted to a smaller collection of equivalent sequences and the relations among them. This abstraction is its induced algebra: a formal pattern that determines which sequences combine, which count as equivalent, and which can be continued. The algebra also identifies which features of step sequences are preserved under the chosen configuration map and equivalence relation. It allows step sequences to be compared within the induced algebra, rather than with reference to the full details of the originating field.

Configurations as Objects

The quotient $\Sigma^+ / \equiv$ introduced above functions as the object layer for the formal constructions developed in subsequent annexes.

Elements of this quotient represent configurations: equivalence classes of step sequences that are treated as the same for purposes of continuation, comparison, and recombination. These equivalence classes serve as the objects on which the formal operations of the annexes are defined.

In Annex I, local sections are compared and assembled relative to this equivalence system. Agreement on overlaps is understood as coincidence at the level of configuration classes.

In Annex II, algebraic composition operates on representatives of these classes and preserves their induced equivalence relations. It is defined so as to respect this equivalence.

In Annex III, geometric relations and alignment measures are defined between configuration spaces whose points correspond to such classes.

Appendix B evaluates methods through their behavior on these configuration equivalence classes under repeated application and refinement.

Appendix C applies the algebraic and sheaf-theoretic operations of the Annexes to a specific instance of systemic failure, validating the diagnostic capacity of the formal objects.

The quotient construction of Annex 0 provides a common object layer that allows local comparison, algebraic composition, geometric orientation, and formal testing to apply to the same underlying configuration.

4. *Semigroup of Methodicity*

The minimal formal object is:

$$(M, \circ) = (\Sigma^+, \text{concatenation})$$

with associativity inherited from concatenation, no identity, no inverses, and quotients defined by equivalence of configurations.

ANNEX I — FUNCTORIAL AND SHEAF-THEORETIC INTEGRATION

I.0 Concept and Rationale

Annex I introduces a sheaf-theoretic formalism for describing agreement and compatibility among locally defined sections over a domain. The constructions developed here concern conditions under which locally defined sections can be related, combined, or extended over overlapping regions.

Throughout this annex, agreement and coincidence among local sections are understood operationally, in the sense of Annex 0; in particular, equalities between restricted sections are to be read as equivalence with respect to the adopted configuration criteria, unless explicitly stated otherwise.

Inquiry proceeds by applying methods on restricted domains U_i , each representing a limited range of conditions, materials, or assumptions. On each patch U_i , applying the method determines a local section s_i over U_i . Where patches overlap, the corresponding sections may be compared. When the local sections, restricted to overlaps, produce equivalent configurations in the sense of Annex 0, they determine a single section over the union of the patches. This agreement condition is the formal criterion for the existence of a global section in Annex I.

Agreement on overlaps may be exact in some cases and approximate in others. In some cases, the local sections coincide; in others, they differ in limited and describable ways.

Whether local sections coincide or differ, in the sense of configuration equivalence, is assessed relative to a given cover.

A cover is a collection of patches whose union represents the portion of a domain currently under inquiry. Inquiry

may proceed by extending, refining, or reorganizing these patches. Further refinements may introduce distinctions that alter whether local sections continue to produce equivalent configurations on overlaps. The openness and extensibility of covers provide a way of representing uncertainty, understood as the persistence of indeterminacy under refinement.

Compatibility may also be examined under successive refinements of covers. When local sections continue to produce equivalent configurations as distinctions are introduced, this persistence shows that the same configuration is maintained under increasing resolution.

When several local sections are available on overlapping patches, one may also consider operations that combine them. Within the sheaf-theoretic formalism, such combination is modeled by a monoidal operation on sections. Different monoidal operations lead to different modes of aggregation and hence to different patterns of compatibility over patches.

I.1 Sites, Presheaves, Sheaves

Sites

Let Dom be a category whose objects are domains or regions U of inquiry and whose morphisms $f : V \rightarrow U$ represent restrictions or refinements. Equip Dom with a Grothendieck topology J , specifying which families $\{U_i \rightarrow U\}$ count as covers.

Presheaf of Methods

Let Struct denote a category of structured sets of methods and admissible transformations between them. A presheaf $\mathcal{M} : \text{Dom}^{op} \rightarrow \text{Struct}$ assigns to each U a collection $\mathcal{M}(U)$ of

methods applicable on U , and to each morphism $f : V \rightarrow U$ a restriction map

$$\rho_{UV} : \mathcal{M}(U) \rightarrow \mathcal{M}(V).$$

Sheaf Condition (Strict)

For a cover $\{U_i \rightarrow U\}$ and sections $s_i \in \mathcal{M}(U_i)$ satisfying

$$\rho_{U_i U_{ij}}(s_i) = \rho_{U_j U_{ij}}(s_j) \quad \text{in } \mathcal{M}(U_{ij}),$$

there exists a unique $s \in \mathcal{M}(U)$ with $\rho_{U U_i}(s) = s_i$.

Local sections that agree on overlaps, in the operational sense fixed in I.0, determine a unique global section over U .

Lax and Fuzzy Descent

In many cases, agreement holds only up to specified transformations α_{ij} . Such situations are modeled by pseudofunctors $\mathcal{M} : \text{Dom}^{op} \rightarrow \text{Cat}$ or stacks, recording both sections and isomorphisms between their restrictions. Descent then requires satisfaction of a cocycle condition for the α_{ij} .

Under this formulation, overlap relations are carried by explicit transition transformations α_{ij} , and the cocycle condition expresses their consistency through triple overlaps.

Čech Viewpoint

For a cover $\mathfrak{U} = \{U_i\}$, a 0-cochain consists of sections (s_i) ; 1-cochains carry transition data on overlaps. A coboundary operator

$$d : C^k(\mathfrak{U}, \mathcal{M}) \rightarrow C^{k+1}(\mathfrak{U}, \mathcal{M})$$

assigns to each family of local data its induced discrepancies on higher overlaps. Cocycles are cochains annihilated by d , expressing compatibility, while coboundaries are those arising

from lower-level data. Where the coefficient structures admit a suitable linearization or abelianization, non-trivial Čech cohomology classes signal obstructions to global sections.

I.1.1 Morphisms Between Domains

Let Dom be the category of domains used in this annex. A morphism $f : U \rightarrow V$ represents a restriction, refinement, or context-shift between domains, and it induces a corresponding restriction map on sections of the presheaf $\mathcal{M}$:

$$f^* : \mathcal{M}(V) \rightarrow \mathcal{M}(U).$$

Morphisms satisfy the standard categorical identities:

$$(g \circ f)^* = f^* \circ g^*, \quad (\text{id}_U)^* = \text{id}_{\mathcal{M}(U)}.$$

When Dom is equipped with a Grothendieck topology J , covers are specified as families $\{U_i \rightarrow U\}$ in Dom . Refinements of covers are represented by morphisms relating one covering family to another.

This fixes the formal basis for restriction maps and for the sheaf and descent conditions used throughout the annex.

I.2 Functorial Behavior and Transposition

Let Meth be the category whose objects are sheaf-theoretic method-assignments $\mathcal{M} : \text{Dom}^{op} \rightarrow \text{Struct}$, and whose morphisms are natural transformations between such assignments.

These morphisms represent structural transpositions in the sense of Part III: controlled relocations of methods into new representational domains that preserve selected relations while discarding or weakening others.

A transposition is represented by a functor

$$T : \text{Meth} \rightarrow \text{Meth}.$$

Such a functor carries method-assignments into new descriptive or operational areas, such as analytic, geometric, combinatorial, or symbolic regimes.

The naturality condition expresses the compatibility of transposition with restriction: applying a transposition and then refining a domain give the same result as refining first and then applying the transposition. This condition formalizes the compatibility of transposition with changes of domain.

Adjoint Transpositions

Some transpositions occur in paired forms, in which one translation moves a method into a more abstract or flexible domain and a corresponding translation carries results back into a more restricted regime. In categorical terms, such pairs may be represented by adjoint functors $L \dashv R$.

In this adjoint formulation, the left adjoint L marks a passage toward symbolic, structural, or generalized representations, while the right adjoint R marks the return of results to the original domain of application. The unit and counit of the adjunction describe what is preserved and what is altered in passing between these representations.

These paired translations formalize the reversible and partially reversible transpositions described in Part III of the Register.

1.2.1 Combination and Intensity

In inquiry, procedures or partial results are often combined to extend or maintain a configuration. In this annex, such acts of combination are represented by equipping each $M(U)$ with a

monoidal product $\otimes$, assumed to be associative and to admit a unit element up to the standard coherence isomorphisms.

The operation $\otimes$ functions as a formal model of joint use, sequential reinforcement, or coordinated application of methods within a domain.

Combination, Intensity, and Implication

In analyses where the cumulative effect of combining local sections is itself under examination, the monoidal product $\otimes$ may be supplemented with an evaluative function. One may introduce a map

$$\| \cdot \| : \mathcal{M}(U) \rightarrow \mathbb{R}_{\geq 0},$$

assigning a nonnegative magnitude to each section.

The behavior of this function under composition—for example, comparison of $\|x\|$, $\|y\|$, and $\|x \otimes y\|$ —provides a way to track how successive or aggregated combinations modify the prominence or persistence of a configuration. Submultiplicative behavior corresponds to diminishing returns under accumulation, while saturating behavior corresponds to thresholds beyond which further combination has limited effect.

Intensity is read off from the comparative behavior of sections under repeated combination within the domain.

When $\mathcal{M}(U)$ is closed monoidal, internal Hom objects $\llbracket x \Rightarrow y \rrbracket$ model relations between sections. These objects characterize how one local configuration extends, regulates possible continuations of, or reorganizes another within the domain. Evaluations or invariants of such objects allow comparative assessment of relational orientation and compatibility among methods.

Descent and Verification

Let $\{U_i \rightarrow U\}$ be a cover and let $s_i \in \mathcal{M}(U_i)$ be local sections.

When the sections s_i , restricted to overlaps, result in equivalent configurations in the sense fixed in I.0, they determine a section $s \in \mathcal{M}(U)$ whose restrictions to each U_i recover the given local sections.

In categorical terms, this expresses the descent condition: local data that agree on overlaps determine global data over the domain.

I.3 Examples and Correspondences

The models introduced in I.0–I.2 apply in formal domains of inquiry. The following examples treat locally applied methods as sections over restricted domains and interpret agreement on overlaps through configuration equivalence in the sense of Annex 0.

Example 1 — Local and Global Proofs

In mathematical inquiry, results are often established under restricted assumptions or within limited regions of applicability. Let U_i represent such a restricted region, defined by boundary conditions or ranges of validity. A proof carried out under these conditions determines a local section s_i over U_i .

When two sets of assumptions overlap, their corresponding proofs may be compared on the shared region. Agreement on overlaps means that the arguments, restricted to common hypotheses, result in equivalent configurations of dependencies and conclusions.

The sheaf condition shows that agreement among overlaps determines a single proof over the union of the domains.

Refinement of hypotheses, reformulation of definitions, or subdivision of cases corresponds to refining the cover. Where agreement has failed, such refinements can alter the overlap relations and permit agreement to be recovered.

This example illustrates how global proofs arise through the assembly of locally valid arguments under conditions of configuration equivalence.

Example 2 — Experimental Design

Consider a family of experimental setups conducted under varying physical, technical, or procedural conditions. Let each setup define a domain U_i . Applying an experimental method within U_i produces a local section s_i , consisting of observations and derived relations.

Overlaps $U_i \cap U_j$ correspond to shared ranges of conditions or protocols. On these overlaps, experimental results may be compared. Agreement means that the locally obtained data, when restricted to common conditions, produce equivalent configurations of relations and measurements.

When agreement holds throughout a cover, the local experimental results assemble into a single description over the combined domain. This expresses that the method functions consistently throughout the covered regimes.

Disagreement isolates dependencies on uncontrolled variables or procedural differences. Refinement of protocols or subdivision of experimental regimes corresponds to refining the cover. Such refinements may restore agreement by separating incompatible regimes.

A protocol that remains compatible under successive refinements can be extended to increasingly broad domains of application.

Summary

In mathematical reasoning, experimental practice, and other formally constructed domains, methods are applied locally within restricted sections. Their results are compared where those sections overlap. Agreement, understood through configuration equivalence, determines whether these local applications can be assembled into a single construction.

Where agreement fails, the failure identifies specific dependencies or incompatible regimes. Refinement of domains and representations provides a way to reorganize inquiry and make assembly possible.

These sheaf-theoretic constructions give a shared formal description of the relation between the local activity of a method and the global organization of its results over a domain.

ANNEX II — N-TUPLE MATRIX

II.0 Motivation

Many methods operate through ordered or layered sequences of procedures, transformations, and checks. As inquiry develops, such sequences are combined, repeated, and compared as the same procedure is applied in different areas.

Annex II introduces an algebraic representation for the composition of multi-stage procedures and interacting sub-methods. The n-tuple matrix shows how multiple layers of a method or method system interact and transform one another under composition.

This representation makes it possible to examine how complex procedures and method systems are assembled from simpler components and how their combined behavior varies with restriction, extension, and repetition.

II.1 Layered Matrix Representation

Each matrix entry is an n-tuple

$$M_{ij} = (m_{ij}^{(1)}, \dots, m_{ij}^{(n)}),$$

where the superscript k indexes a fixed layer of the construction. The indices i, j range over the positions of the matrix.

Each layer k carries its own composition rule, specified in Section II.2, and the components $m_{ij}^{(k)}$ are composed only within their respective layers.

The layered format permits different aspects of a procedure to be combined in parallel while remaining formally distinct.

II.2 Algebraic Consistency Conditions

Composition is defined layerwise. For two matrices M and N , the composed matrix $M \otimes N$ has entries

$$(M \otimes N)_{ij}^{(k)} = \text{Agg}_k \left(f_k(m_{i\ell}^{(k)}, n_{\ell j}^{(k)}) \right)_\ell,$$

where each f_k is a binary operation acting on layer k .

Each operation f_k is assumed to be associative wherever the composition is defined.

The operator Agg_k denotes the aggregation over the intermediate index ℓ appropriate to layer k , for example summation, integration, or maximization, depending on how that layer combines contributions.

Repeated application of stages is represented by iterated composition using $\otimes$.

Alignment measures may be defined on individual layers and combined to obtain a global evaluation of the composite procedure.

II.2.1 Layerwise Conditions

For the layerwise composition defined in II.2 to function as an algebraic operation on methods in combination, each layer k must satisfy the following conditions.

1. **Associativity.** For entries a, b, c in layer k ,

$$f_k(f_k(a, b), c) = f_k(a, f_k(b, c)).$$

2. **Identity Element.** There exists an element e_k in layer k such that, for all admissible a ,

$$f_k(e_k, a) = a = f_k(a, e_k).$$

The aggregation operator Agg_k is assumed to admit neutral families $\{\delta_{i\ell}^{(k)}\}$ such that, when $m_{i\ell}^{(k)} = e_k \delta_{i\ell}^{(k)}$, the corresponding matrix acts as an identity for $\otimes$ on layer k .

3. **Layerwise Independence.** For any two matrices M, N , and for all indices i, j and layers k ,

$$(M \otimes N)_{ij}^{(k)} = \text{Agg}_k \left(f_k(m_{i\ell}^{(k)}, n_{\ell j}^{(k)}) \right)_\ell,$$

and depends only on the k -th components of M and N , with neither f_k nor Agg_k referencing components from layers $k' \neq k$.

Each component $(M \otimes N)_{ij}^{(k)}$ is obtained from the k -th components of M and N by the operation f_k and the aggregation Agg_k .

These conditions ensure that composite procedures can be examined and computed layer by layer, and that interactions between layers arise only through explicitly introduced couplings. They also make it possible to compare different ways of combining the same components within a method system.

II.3 Transition Maps

A restriction map

$$r(U \rightarrow V) : \mathcal{M}(U) \rightarrow \mathcal{M}(V)$$

represents the passage from a broader domain U to a narrower domain V .

Such maps make explicit which aspects of a local application are preserved under this passage and which are altered or omitted. They provide the formal basis for comparing applications of a method and for analyzing transpositions as described in Part III of the Register.

Transition maps specify how locally applied methods persist, vary, or separate under restriction and reformulation.

II.3.1 Compatibility With Sheaf-Theoretic Restriction

For the n -tuple matrix formalism to be compatible with the restriction maps defined in Annex I, every transition map $r(U \rightarrow V) : \mathcal{M}(U) \rightarrow \mathcal{M}(V)$ must satisfy the following conditions on each matrix layer.

Here the objects are domains and the morphisms are restriction maps as defined in Annex I.

Formally, if $M \in \mathcal{M}(U)$ is an n -tuple matrix over domain U , then the restricted matrix $r(M) \in \mathcal{M}(V)$ must satisfy:

1. **Layerwise Restriction.** For each layer k ,

$$r(M)^{(k)} = r(M^{(k)}).$$

2. **Compatibility With Composition.** For any two matrices M, N defined on U ,

$$r(M \otimes N) = r(M) \otimes r(N),$$

whenever the composition is defined.

3. **Functoriality of Restriction.** For composable domain morphisms $U \rightarrow V \rightarrow W$,

$$r_{U \rightarrow W} = r_{V \rightarrow W} \circ r_{U \rightarrow V}.$$

These conditions ensure that algebraic composition in Annex II is consistent with the restriction and gluing conditions of Annex I.

II.3.2 Cross-Annex Coherence Clause

For consistency between Annexes I–III, every transition map $r(U \rightarrow V)$ must interact compatibly with the geometric relations defined in Annex III.

Let $\Sigma_{ij,i'j'}^{(U)}$ be an interface between configuration spaces associated with domain U , and let $\Sigma_{ij,i'j'}^{(V)}$ denote the corresponding interface after restriction to V , whenever such a correspondence is defined.

Let $\theta_{ij,i'j'}^{(U)}$ and $\theta_{ij,i'j'}^{(V)}$ denote the associated interface divergence parameters.

Compatibility requires that

$$\cos(\theta_{ij,i'j'}^{(V)}) \leq \cos(\theta_{ij,i'j'}^{(U)}),$$

whenever both quantities are defined.

This condition expresses that restriction to a narrower domain cannot introduce improved geometric alignment between interfaces that was not already present on the broader domain.

II.4 Worked Examples

The following examples make the operations of composition, aggregation, restriction, and transposition explicit. They show how layered matrices representing interacting procedures are constructed and manipulated under the rules of Sections II.2–II.3.

Example A: Two-layer composition in a 3×3 matrix

This example fixes a small layered matrix and computes an explicit entry of a composite.

Assume two layers: a conceptual layer (c) and a performative layer (p), corresponding to two instances of the abstract layer index k .

Let M be a 3×3 n-tuple matrix with entries

$$M_{ij} = (o_{ij}^c, o_{ij}^p).$$

Write

$$M = \begin{pmatrix} (o_{11}^c, o_{11}^p) & (o_{12}^c, o_{12}^p) & (o_{13}^c, o_{13}^p) \\ (o_{21}^c, o_{21}^p) & (o_{22}^c, o_{22}^p) & (o_{23}^c, o_{23}^p) \\ (o_{31}^c, o_{31}^p) & (o_{32}^c, o_{32}^p) & (o_{33}^c, o_{33}^p) \end{pmatrix}.$$

Fix layerwise operations by

$$f_c(a, b) = a + b, \quad f_p(a, b) = ab,$$

and choose aggregation in both layers by summation over the intermediate index,

$$\text{Agg}_c(\cdot)_\ell = \sum_{\ell=1}^3 (\cdot), \quad \text{Agg}_p(\cdot)_\ell = \sum_{\ell=1}^3 (\cdot).$$

Then the composite $M \otimes M$ has layerwise entries

$$(M \otimes M)_{ij}^{(c)} = \sum_{\ell=1}^3 (o_{i\ell}^c + o_{\ell j}^c), \quad (M \otimes M)_{ij}^{(p)} = \sum_{\ell=1}^3 (o_{i\ell}^p o_{\ell j}^p).$$

These choices presuppose closure and distributive interaction between aggregation and layerwise operations, in the sense that Agg_k preserves finite combinations formed by f_k whenever the relevant expressions are defined.

For example, the $(1, 2)$ entry of $M \otimes M$ is the ordered pair

$$(M \otimes M)_{12} = \left(\sum_{\ell=1}^3 (o_{1\ell}^c + o_{\ell 2}^c), \sum_{\ell=1}^3 (o_{1\ell}^p o_{\ell 2}^p) \right).$$

The conceptual component expands as

$$(M \otimes M)_{12}^{(c)} = (o_{11}^c + o_{12}^c) + (o_{12}^c + o_{22}^c) + (o_{13}^c + o_{32}^c).$$

The conceptual layer accumulates by additive interaction throughout intermediate stages, while the performative layer accumulates by multiplicative interaction through the same intermediates. Other choices of Agg_k (for example, maximization or averaging) alter the manner in which intermediate contributions are accumulated without changing the layerwise separation of operations.

Example B: Comparison of Applications Under Transposition via Transition Maps

This example illustrates how transition maps provide a formal basis for comparing applications of a method between domains and for analyzing their transpositions, as described in Section II.3. It shows how restriction, composition, and transposition remain fully compatible.

We consider a classical analytic-to-geometric transposition from differential geometry: the integration of vector fields to their associated flows.

Let U be a domain in a smooth manifold and let $V \subseteq U$ be a subdomain. Suppose that on each domain we are given local evolution rules described by vector fields.

For each matrix position (i, j) , define the k -th layer by

$$m_{ij}^{(k)} = X_{ij}^{(U)},$$

where $X_{ij}^{(U)}$ is a vector field defined on U .

Restriction to V induces vector fields

$$X_{ij}^{(V)} = r(X_{ij}^{(U)}),$$

and hence a restricted matrix

$$r(M)_{ij}^{(k)} = X_{ij}^{(V)}.$$

Define layerwise composition by

$$f_k(X, Y) = X + Y, \quad \text{Agg}_k(\cdot)_\ell = \sum_\ell (\cdot).$$

Then

$$(M \otimes N)_{ij}^{(k)} = \sum_\ell (X_{i\ell}^{(U)} + Y_{\ell j}^{(U)}),$$

and compatibility with restriction yields

$$r(M \otimes N) = r(M) \otimes r(N).$$

Composite evolution rules may be compared on U and V at the level of infinitesimal generators.

Classically, each vector field X determines a flow $\Phi_t = \exp(tX)$ satisfying

$$\Phi_0 = \text{id}, \quad \Phi_{t+s} = \Phi_t \circ \Phi_s.$$

Define a transposition operator T on the k -th layer by

$$T(X_{ij}) = \Phi_t^{ij},$$

where Φ_t^{ij} is the flow generated by X_{ij} .

Applying T entrywise yields

$$T(M)_{ij} = (m_{ij}^{(1)}, \dots, \Phi_t^{ij}, \dots, m_{ij}^{(n)}).$$

Restriction of vector fields induces restriction of flows, and satisfies

$$T(r(M)) = r(T(M)).$$

We may now compare two applications of the same method on U and V .

On U , the transposed matrix $T(M)$ encodes global evolution operators Φ_t^{ij} . On V , the transposed matrix $T(r(M))$ encodes the restricted operators $\Phi_t^{ij}|_V$.

Because transposition commutes with restriction, these operators satisfy

$$r(T(M)) = T(r(M)).$$

Consequently, comparison of applications on U and V may be carried out either:

- at the analytic level, by comparing $X_{ij}^{(U)}$ and $X_{ij}^{(V)}$, or
- at the geometric level, by comparing Φ_t^{ij} and $\Phi_t^{ij}|_V$,

with equivalent results.

Transition maps mediate comparison of applications between domains and ensure that analytic and geometric representations of a method remain aligned under transposition.

Annex II provides the algebraic syntax underlying the geometric semantics of Annex III.

ANNEX III — MATRIX OF CONFIGURATION SPACES

This annex develops a geometric representation of composed and restricted methods by organizing configuration spaces into a matrix linked by boundary identifications and transition maps.

Throughout this annex, the symbols M_{ij} denote configuration spaces indexed by matrix position. They are distinct from the n -tuple matrices M used in Annex II, which represent algebraic compositions of methods.

III.0 Rationale

Annex III introduces a geometric representation for the relational organization of methods. Arranging configuration spaces in a matrix exposes how different dimensions of a method interact through shared boundaries and transition rules.

Let I and J be index sets. The indices $i \in I$ and $j \in J$ label positions in the matrix and correspond to distinct procedural dimensions of inquiry.

Adjacent entries are related by explicit identifications, which determine how configurations are carried from one position to another.

This construction provides a geometric counterpart to the algebraic compositions of Annex II and the sheaf-theoretic relations of Annex I.

III.1 Definition

Each entry M_{ij} is a configuration space: a domain of available configurations equipped with specified boundary relations and

transition maps to neighboring entries.

For each M_{ij} , let ∂M_{ij} denote the collection of boundary configurations through which transitions to adjacent entries are defined. These boundaries determine how configurations may pass from one position in the matrix to another.

For adjacent entries M_{ij} and $M_{i'j'}$, the shared interface $\Sigma_{ij,i'j'}$ is understood as a specified subset of ∂M_{ij} and $\partial M_{i'j'}$ on which boundary identifications are defined.

Boundary Compatibility

For adjacent entries M_{ij} and $M_{i'j'}$, let

$$\iota_{ij} : \partial M_{ij} \hookrightarrow M_{ij}$$

denote the inclusion of boundary configurations, and let

$$\phi_{ij,i'j'} : \partial M_{ij} \rightarrow \partial M_{i'j'}$$

denote the identification map relating corresponding boundaries.

Compatibility requires that for each $x \in \Sigma_{ij,i'j'}$,

$$\iota_{ij}(x) \sim \iota_{i'j'}(\phi_{ij,i'j'}(x)),$$

where $\sim$ denotes the equivalence relation induced by the boundary identifications.

This condition ensures that configurations transferred between adjacent entries remain comparable and that interface transitions do not introduce discontinuities or contradictions.

III.2 Global Organization

Let $\bigsqcup_{(i,j) \in I \times J} M_{ij}$ denote the disjoint union of the configuration spaces. Let $\mathcal{E}$ denote the collection of all shared boundaries

$\Sigma_{ij,i'j'}$ together with their identification maps $\phi_{ij,i'j'}$. These identifications equip $\mathcal{E}$ with a system of relations that connect adjacent entries and allow configurations to be compared and transferred throughout the matrix.

The matrix $\{M_{ij}\}_{i \in I, j \in J}$ forms an indexed family whose entries are linked by specified transition rules along both indices.

A global section of this family consists of a choice of configurations $x_{ij} \in M_{ij}$ for each position (i, j) such that all boundary identifications are respected. That is, whenever two entries are adjacent, the boundary data induced by the chosen configurations agree under the corresponding identification maps along the shared interface.

Such families represent configurations whose local sections satisfy the equivalence condition on overlaps (Annex I, Descent and Verification) and correspond to the global sections defined in Annex I.

III.3 Geometric Alignment Measure

Along each shared boundary $\Sigma_{ij,i'j'}$ between adjacent configuration spaces M_{ij} and $M_{i'j'}$, configurations are related by the boundary identification map $\phi_{ij,i'j'}$.

In this annex, a geometric alignment measure is used to represent how closely the transition rules on the two sides of an interface agree in their treatment of boundary configurations. This measure models alignment through boundary relations and does not replace the general notion of alignment developed elsewhere in the Register.

Alignment at an interface expresses the degree to which configurations transferred throughout $\Sigma_{ij,i'j'}$ preserve their relational role within the respective spaces.

To represent this quantitatively, associate to each interface

a parameter $\theta_{ij,i'j'}$, interpreted as a measure of divergence between the transition structures on the two sides and corresponding to the alignment parameters used in Annex II. Smaller values correspond to closer agreement; larger values correspond to greater divergence.

Recall that $\mathcal{E}$ denotes the collection of all shared boundaries $\Sigma_{ij,i'j'}$ together with their identification maps. For each interface $e \in \mathcal{E}$, let w_e denote its weight and θ_e its divergence parameter.

For the matrix of configuration spaces $\mathcal{M} = \{M_{ij}\}_{(i,j) \in I \times J}$, define the geometric alignment value by

$$A_{\text{geom}}[\mathcal{M}] = \frac{\sum_{e \in \mathcal{E}} w_e \cos \theta_e}{\sum_{e \in \mathcal{E}} w_e}.$$

This quantity aggregates interface-level agreement into a single measure of how consistently configurations are related throughout the matrix.

Higher values indicate close agreement of boundary transitions. Lower values indicate systematic divergence between adjacent configuration spaces.

III.4 Sheaf-Theoretic View

The family of configuration spaces $\{M_{ij}\}_{(i,j) \in I \times J}$, together with the boundary identification maps $\phi_{ij,i'j'}$, determines a presheaf $\mathcal{F}$ with values $\mathcal{F}_{ij}$.

This presheaf is defined over the adjacency and boundary-identification system of the matrix, rather than over the domain category Dom used in Annex I.

For each entry M_{ij} , the presheaf $\mathcal{F}_{ij}$ assigns the set of local configurations and transition data defined on that entry.

Each boundary identification

$$\phi_{ij,i'j'} : \partial M_{ij} \rightarrow \partial M_{i'j'}$$

induces a restriction map

$$\rho_{ij,i'j'} : \mathcal{F}_{ij} \rightarrow \mathcal{F}_{i'j'},$$

expressing how configurations are transferred between adjacent entries.

The collection $(\{\mathcal{F}_{ij}\}, \rho)$ is required to satisfy a descent condition: whenever sections defined on a family of adjacent entries agree under all induced restriction maps, there exists a section on their union whose restrictions recover the given local sections.

Agreement along shared boundaries is sufficient for extension to larger submatrices. Local compatibility determines global assembly.

The organization of configuration spaces in the matrix is governed by the same local-to-global principle introduced in Annex I, now applied to boundary-mediated relations between entries.

This formulation makes explicit how geometric relations in the matrix are integrated with sheaf-theoretic conditions on agreement and extension.

III.5 Development and Revision

Let $\mathcal{M}_t$ denote the matrix of configuration spaces at stage t of an inquiry, where t indexes successive applications, refinements, or reorganizations of a method.

A transition from stage t to stage $t + 1$ is represented by a morphism

$$T_t : \mathcal{M}_t \rightarrow \mathcal{M}_{t+1},$$

which maps each configuration space $M_{ij}^{(t)}$ to a corresponding

space $M_{ij}^{(t+1)}$ and carries boundary identifications and transition relations forward in a manner compatible with the boundary identifications, transition maps, and alignment relations defined in this annex.

Such morphisms correspond to reformulations of representations or extensions of domain.

Compatibility with Annexes I and II requires that each T_t respect restriction maps and algebraic composition, so that local comparisons and compositional relations remain traceable through stages.

A sequence

$$\mathcal{M}_0 \xrightarrow{T_0} \mathcal{M}_1 \xrightarrow{T_1} \mathcal{M}_2 \xrightarrow{T_2} \dots$$

represents the successive development of a method through repeated application and revision.

Configuration relations persist under continued use and refinement, when successive morphisms preserve boundary identifications and alignment measures within bounded variation.

When successive morphisms introduce changes in boundary relations, transition maps, or alignment values, configuration relations are reorganized, corresponding to modification or divergence of application.

III.6 Example — Spherical Mirror Configuration Space

The spherical mirror introduced in the Register (Part IV) provides a well-defined instance of a matrix of configuration spaces governed by explicit geometric rules.

In this construction, reflected images are generated by fixed relations determined by the curvature and orientation of a

virtual spherical surface. Spherical inversion serves as an idealized representation of the reflection geometry. The resulting reflection space constitutes a configuration space in the sense of this annex.

Let the mirror be a sphere of radius R , and let physical points $p \in \mathbb{R}^3 \setminus \{0\}$ be mapped to virtual points by inversion

$$I_R(p) = \frac{R^2}{\|p\|^2} p.$$

This map is smooth away from the origin, conformal, and involutive.

The image of physical space under I_R determines the virtual manifold $V = \mathbb{R}^3 \setminus \{0\}$, equipped with the pulled-back metric

$$g_{ab}^{(v)}(p) = \Omega(p)^2 \delta_{ab}, \quad \Omega(p) = \frac{R^2}{\|p\|^2}.$$

Each entry M_{ij} represents a restricted region of the virtual manifold, specified by bounded ranges of curvature, viewing position, and reflection parameters. Within such a region, admissible configurations consist of image relations generated by the corresponding reflection rules.

Boundaries ∂M_{ij} correspond to limiting regimes in which reflection behavior changes, such as transitions between convex and near-planar regions, or between distinct viewing angles.

Boundary identification maps

$$\phi_{ij,i'j'} : \partial M_{ij} \rightarrow \partial M_{i'j'}$$

relate reflection patterns that coincide at shared parameter limits. These identifications ensure that images produced in adjacent regions may be compared and transferred without ambiguity.

Alignment between directions is measured using the virtual metric. For tangent vectors

$$v \in T_p M_{ij}, \quad w \in T_p M_{i'j'},$$

define

$$A^{(v)}(v, w) = \frac{\langle v, w \rangle_{g^{(v)}}}{\|v\|_{g^{(v)}} \|w\|_{g^{(v)}}}.$$

Because $g^{(v)}$ is conformal to the Euclidean metric, angular relations are preserved under inversion, while magnitudes are scaled by the conformal factor Ω .

Interface Divergence

Let two adjacent cells M_{ij} and $M_{i'j'}$ share an interface

$$\Sigma_{ij,i'j'} \subset \partial M_{ij} \cap \partial M_{i'j'}.$$

At a point $p \in \Sigma_{ij,i'j'}$, choose representative tangent directions

$$v_{ij} \in T_p M_{ij}, \quad v_{i'j'} \in T_p M_{i'j'}.$$

Define the interface divergence parameter

$$\theta_{ij,i'j'}(p) = \arccos \left(\frac{\langle v_{ij}, v_{i'j'} \rangle_{g^{(v)}}}{\|v_{ij}\|_{g^{(v)}} \|v_{i'j'}\|_{g^{(v)}}} \right).$$

Because inversion is conformal, angular comparison between interfaces depends only on Euclidean angles. Interface misalignment in this configuration space is determined by directional divergence rather than metric magnitude.

Alignment Variation and Curvature

The gradient of the conformal factor is

$$\nabla \Omega(p) = -2R^2 \frac{p}{\|p\|^4}.$$

Consequently, alignment variation is governed by radial changes in Ω . Near the mirror center, where $\|p\|$ is small, alignment gradients are large. Far from the center, alignment gradients decay.

Spatial variation of alignment in this configuration space is controlled entirely by the radial behavior of the conformal scaling.

Restriction and Monotonicity

Let $V' \subset V$ be a restricted domain defined by

$$r^{(v)} \geq r_0, \quad r^{(v)} = \|I_R(p)\| = \frac{R^2}{\|p\|}.$$

On V' , the conformal factor Ω is bounded and its gradient magnitude decreases monotonically with increasing radius. For interfaces contained in V' ,

$$\cos(\theta_{ij,i'j'}^{(V')}(p)) = \cos(\theta_{ij,i'j'}^{(V)}(p)),$$

whenever both quantities are defined and the same representative tangent directions are used.

Restriction to narrower radial regimes does not by itself alter geometric alignment; any change in alignment must arise from a change in the selected interfaces or in the representative tangent directions used for comparison.

Continuation Between Cells

Let γ be a curve crossing from M_{ij} to $M_{i'j'}$ at a point $p \in \Sigma_{ij,i'j'}$.

Geometric continuation within the interface requires:

1. Compatibility of tangent directions with the boundary identification,
2. Bounded interface divergence,

3. Admissibility of the transported tangent in the target cell.

Because inversion is smooth and conformal away from the origin, continuation preserves angular direction while rescaling tangent magnitudes by the factor Ω .

In this configuration space, continuation between cells preserves direction but alters magnitude in accordance with the conformal structure.

Successive Refinement

Successive modifications of mirror parameters generate a sequence

$$\mathcal{M}_0 \rightarrow \mathcal{M}_1 \rightarrow \mathcal{M}_2 \rightarrow \cdots$$

of configuration matrices, representing progressive refinement or reorganization of the virtual reflection space.

Under such refinements, local reflection behavior persists, while alignment gradients, interface divergences, and continuation properties vary according to the induced changes in the conformal factor and cell boundaries.

Summary

In the spherical mirror, configuration spaces are determined by inversion in a sphere. Boundary relations, alignment measures, restriction maps, and continuation follow from the same mapping. Interfaces inherit their divergence from the conformal factor. Restriction corresponds to limiting the domain of inversion. Continuation is governed by the same conformal relations.

All of these arise from the single transformation that defines the space.

The Matrix of Configuration Spaces is a geometric model linking algebra (Annex II) and sheaf logic (Annex I).

APPENDIX A — THE ATLAS

A.0 Purpose and Status of the Atlas

The Atlas is a cartographic interface to the Register. It is intended to display, in a compact and comparable form, the principal descriptive features of methods as they appear in practice.

Its function is comparative. Within it, methods are placed side by side and examined under shared descriptive dimensions.

All coordinates and entries are assigned by an observer for the purposes of description and analysis.

A.1 Methods as Atlas Entries

Let $M = \{m_1, \dots, m_p\}$ be a finite collection of methods under consideration.

Each method is represented in the Atlas by a descriptive tuple

$$m = (G, B, A, V, I, H)$$

where the components are primary functional aspects of the method, as defined in the Register.

These components are:

- **G (Gesture):** The characteristic mode of operation or activity.
- **B (Beacon):** The form of reference, evidence, or guidance selected.
- **A (Alignment):** The degree of fit and orientation among steps within a configuration.

- **V (Veracity):** The extent to which a configuration remains oriented towards purpose.
- **I (Intensity):** The level and persistence of engagement.
- **H (Uncertainty):** The acknowledged scope of indeterminacy.

Each component is interpreted as a fixed descriptive reference for comparative purposes. Interpretation proceeds through qualitative description rather than numerical scaling.

A.2 The Atlas Matrix

The Atlas is represented as a matrix

$$\mathcal{A} = (a_{ij})$$

whose rows correspond to methods and whose columns correspond to descriptive dimensions.

Each row of the Atlas matrix consists of the six coordinates (G, B, A, V, I, H) associated with the corresponding method.

The entry a_{ij} holds the description assigned to method m_i along dimension j .

A.3 Qualitative Gradients

Several of the coordinates may be described along qualitative gradients. Such gradients can assist in locating methods within their ranges of activity.

For example, alignment may range from *fragmented* to *integrated*, veracity from *informal* to *formally validated*, intensity from *episodic* to *sustained*, and uncertainty from *narrow* to *extensive*.

A.4 Domain Regions and Method Families

In addition to coordinate dimensions, methods may be grouped into regions that reflect their primary uses.

Principal regions developed in the Register include:

- **Operational:** Procedural, stepwise, or combinational.
- **Systemic:** Interdependent, multi-component methods.
- **Geometric:** Spatial, topological, or manifold-based.
- **Virtual:** Constructed or simulated environments.
- **Formal:** Symbolic, algebraic, or rule-based systems.
- **Evaluative:** Assessment of adequacy, fitness, and veracity.

A.5 Use of the Atlas

The Atlas is designed for the comparison of methods along selected descriptive dimensions. Similarities and patterns of development become visible within a shared representation.

Atlas entries may be reorganized by reordering rows or columns, selecting submatrices, transposing the matrix orientation, and juxtaposing alternative descriptions. Through transposition, the axis of inquiry is rotated: descriptive dimensions become the primary objects, and methods become their instances. This orthogonal view is generative: it exposes relations among dimensions that are independent of any single method, allowing the observer to identify dependencies that otherwise remain hidden.

A.6 Illustrative Atlas Matrix

The following example illustrates the use of the Atlas representation for a small collection of methods developed in the Register.

Four methods are considered: the Plausibility Machine, the Spherical Mirror, the n-Tuple Matrix, and the Fuzzy Causal Set. A single descriptive orientation is adopted. Alignment is described using the scale *weak, moderate, strong*. Veracity is described using the scale *informal, semi-formal, formal*. Intensity is described using the scale *low, moderate, high*. Uncertainty is described using the scale *narrow, moderate, broad*.

Method	G (Gesture)	B (Beacon)	A (Alignment)	V (Veracity)	I (Intensity)	H (Uncertainty)
Plausibility Machine	Trial Configuration	Plausibility	Moderate	Informal	High	Broad
Spherical Mirror	Observation	Resonance	Moderate	Semi-formal	Moderate	Moderate
n-Tuple Matrix	Transformation	Alignment	Strong	Formal	High	Narrow
Fuzzy Causal Set	Juxtaposition	Plausibility	Moderate	Semi-formal	Moderate	Broad

This matrix frames a set of coordinate descriptions assigned under a common descriptive orientation. Different choices of scale or assignment would produce different Atlas constructions.

Interpretation of the Example

The matrix permits several comparative features to be examined directly.

Consider first the **Alignment** column. Three of the four methods are described as exhibiting *moderate* alignment, while the n-Tuple Matrix is described as exhibiting *strong* alignment. This distinguishes the n-Tuple Matrix as more tightly coordinated in its internal organization.

The **Veracity** column displays a graded progression from *informal* through *semi-formal* to *formal*. The Plausibility Machine occupies the informal position, the Spherical Mirror and the Fuzzy Causal Set occupy intermediate positions, and the n-Tuple Matrix occupies the formal position.

Comparison of the **Beacon** and **Uncertainty** columns reveals a further relation. The two methods whose primary beacon is *plausibility* are also associated with *broad* uncertainty, whereas the method whose beacon is *alignment* is associated with *narrow* uncertainty.

A.7 Transposition

The generative capacity of the Atlas appears when the matrix is transposed. By rotating the arrangement, the observer shifts the object of inquiry. Descriptive dimensions become the rows, and methods become the columns.

The transposed matrix $\mathcal{A}^T$ takes the following form:

Dimension	Plausibility Machine	Spherical Mirror	n-Tuple Matrix	Fuzzy Causal Set
G	Trial Configuration	Observation	Transformation	Juxtaposition
B	Plausibility	Resonance	Alignment	Plausibility
A	Moderate	Moderate	Strong	Moderate
V	Informal	Semi-formal	Formal	Semi-formal
I	High	Moderate	High	Moderate
H	Broad	Moderate	Narrow	Broad

Interpretation of the Transposition

The transposed view displays the distribution of each attribute throughout the current field of inquiry.

Reading the row for **Intensity (I)** shows that high intensity appears in both the most informal method (Plausibility Machine) and the most formal one (n-Tuple Matrix). This indicates that intensity is a property independent of formal rigor.

Reading the row for **Uncertainty (H)** displays the range of indeterminacy present in the collection, from 'narrow' to 'broad'.

Through rotation, the Atlas allows the observer to scan the dimensions themselves, checking for correlations or gaps in the set of methods under study.

The Atlas makes comparison possible through explicit arrangement. The matrix makes visible clustering tendencies and correlations among descriptive dimensions.

APPENDIX B — TESTING

B.0 Purpose

This appendix outlines a testing process for generative methods.

A method M is evaluated through its performance on a domain D . Testing evaluates whether a method is aligned with its purpose and maintains veracity within the domain in which it is applied.

B.1 Objects and Notation

All quantities introduced in this appendix are defined relative to a specified evaluative regime, including a domain, an interface, and a reference behavior. Their numerical form serves to make compatibility or breakdown explicit under testing conditions.

Evaluation presupposes that the method is applied with sufficient intensity to produce a distinguishable configuration at the interface. Where intensity falls below a minimal detection threshold, the descriptive quantities a_e, v_e, u_e are undefined.

Let M denote a method and D a domain of application. Let $\mathcal{E}(M, D)$ be a finite or countable set of evaluative interfaces e , each specifying a situation in which M is applied to D .

For each interface $e \in \mathcal{E}(M, D)$, let $y_e(M)$ denote the output or behavior of M at e , and let y_e^{ref} denote the corresponding reference behavior or constraint appropriate to that interface.

B.2 Local Tests

Local tests evaluate a method at individual interfaces.

B.2.1 Local Alignment, Veracity, and Uncertainty Conditions

For an interface e , define:

$$a_e(M) := A(y_e(M), y_e^{\text{ref}})$$

At interface e , the alignment score $a_e(M)$ is defined relative to the reference behavior specified by the evaluative regime.

$$v_e(M) := V(y_e(M))$$

For the evaluative regime under consideration, $v_e(M)$ indicates whether the method's output satisfies the reference behavior at interface e .

$$u_e(M) := U(y_e(M))$$

The value $u_e(M)$ quantifies indeterminacy encountered at interface e under repetition or perturbation of the method.

At an interface e , the following alignment, veracity, and uncertainty conditions hold for the method M , with the local scores $a_e(M)$, $v_e(M)$, and $u_e(M)$ taken to be normalized to the interval $[0, 1]$ under the evaluative regime.

1. Alignment condition

$$a_e(M) \geq a_{\min},$$

for some threshold $a_{\min} \in (0, 1)$.

2. Veracity condition

$$v_e(M) \geq v_{\min},$$

for some threshold $v_{\min} \in (0, 1)$.

3. Uncertainty bound

$$u_e(M) \leq u_{\max},$$

where $u_{\max} \in (0, 1)$ is fixed by the testing regime.

These thresholds are specified as part of the testing specification and are derived from domain constraints and practice-specific considerations.

Threshold values are selected parameters of an evaluative regime. Altering them corresponds to changing tolerance for misalignment or approximation within a domain.

B.2.2 Local Test Operator

Define the local test operator:

$$T_{\text{loc}}(M, D, e) = \begin{cases} 1 & \text{if all local alignment, veracity, and} \\ & \text{uncertainty conditions are satisfied at} \\ & e, \\ 0 & \text{otherwise.} \end{cases}$$

T_{loc} establishes the outcome of local testing at interface e ; it is an instrumented test result.

A method passes local testing on D when:

$$T_{\text{loc}}(M, D, e) = 1 \quad \forall e \in \mathcal{E}_{\text{local}}(M, D),$$

where $\mathcal{E}_{\text{local}}(M, D) \subseteq \mathcal{E}(M, D)$ is the selected set of local test interfaces.

B.3 Composite Tests

Composite tests evaluate compatibility under interaction among parts of a method.

Let $M = (\mathcal{C}_1, \dots, \mathcal{C}_k)$ denote a compound method with identifiable components.

B.3.1 Composite Interaction Conditions

Composite interaction is examined through:

- **Mutual alignment**, measuring whether components are similarly oriented when applied to shared interfaces.
- **Scale compatibility**, assessing whether components operate on commensurate levels of granularity and intensity.
- **Iterative continuation**, examining changes in alignment and compatibility when components interact over sequences of applications.

B.3.2 Composite Test Operator

Define:

$$T_{\text{comp}}(M, D) = \begin{cases} 1 & \text{if composite interaction conditions are satisfied,} \\ 0 & \text{otherwise.} \end{cases}$$

Passing composite tests indicates that components remain compatible as a system under interaction on the domain.

B.4 Global Alignment Functional

The global alignment functional is evaluative in its application and normative only relative to the thresholds specified by a testing regime.

Local and composite tests are necessary but not sufficient for generative validity. A global measure integrates behavior throughout the domain.

Partition $\mathcal{E}(M, D)$ into:

$$\mathcal{E}_{\text{local}}(M, D), \quad \mathcal{E}_{\text{comp}}(M, D), \quad \mathcal{E}_{\text{eval}}(M, D).$$

The evaluation set $\mathcal{E}_{\text{eval}}(M, D)$ is chosen to sample the operational range of the domain.

B.4.1 Definition

Define the global alignment functional:

$$\mathcal{C}[M \mid D] = \frac{1}{|\mathcal{E}_{\text{eval}}(M, D)|} \sum_{e \in \mathcal{E}_{\text{eval}}(M, D)} f(a_e(M), v_e(M), u_e(M)),$$

where

$$f(a, v, u) = \lambda_A a + \lambda_V v - \lambda_U u,$$

with non-negative weights $\lambda_A, \lambda_V, \lambda_U$ fixed by the evaluative regime for the domain.

B.4.2 Threshold Regimes

For thresholds $\tau_{\text{low}} < \tau_{\text{high}}$:

- M passes the global alignment threshold if $\mathcal{C}[M \mid D] \geq \tau_{\text{high}}$.
- M is marginal if $\tau_{\text{low}} \leq \mathcal{C}[M \mid D] < \tau_{\text{high}}$.
- M fails the global alignment threshold if $\mathcal{C}[M \mid D] < \tau_{\text{low}}$.

B.5 Generative Validity

A method is generatively valid when local tests, composite tests, and the global alignment condition are satisfied simultaneously.

B.5.1 Definition

A method M is generatively valid on domain D if:

1. All local tests are satisfied:

$$T_{\text{loc}}(M, D, e) = 1 \quad \forall e \in \mathcal{E}_{\text{local}}(M, D),$$

2. Composite tests are satisfied:

$$T_{\text{comp}}(M, D) = 1,$$

3. Global alignment exceeds the high threshold:

$$\mathcal{C}[M \mid D] \geq \tau_{\text{high}}.$$

B.5.2 Iterative Continuation

Let M_t denote the method after t cycles of use or refinement. Iterative continuation is evaluated by examining $\mathcal{C}[M_t \mid D]$ over successive iterations.

A method exhibits generative continuation if there exists a horizon T such that for all $t \geq T$,

$$\mathcal{C}[M_t \mid D] \geq \tau_{\text{high}},$$

and local alignment, veracity, and uncertainty conditions and composite test conditions are satisfied over the same horizon.

Generative validity is provisional. A method may satisfy all current tests yet fail under extension, refinement, or application to a broader evaluative interface. Validity signifies sustained performance under testing.

B.6 Diagnostic Patterns

This section describes characteristic diagnostic patterns in terms of $\mathcal{C}$ and its components.

- **Drift:** A persistent decrease in $\mathcal{C}[M_t \mid D]$ over successive evaluation cycles.
- **Overfit:** $\mathcal{C}[M \mid D]$ is high on a restricted evaluation set but decreases when the evaluation set is broadened or diversified.
- **Underfit:** $\mathcal{C}[M \mid D]$ remains below the high threshold throughout refinement.
- **Systemic Low Alignment:** Many evaluation interfaces exhibit consistently low alignment values, so that $\mathcal{C}[M \mid D]$ falls below the global alignment thresholds.
- **Collapse:** Local alignment, veracity, and uncertainty conditions and composite test conditions cease to be satisfied among a significant portion of evaluation interfaces, accompanied by large fluctuations in $\mathcal{C}[M \mid D]$.

These patterns serve as diagnostic indicators within the testing regime.

B.7 Corrective Measures

Corrective measures are defined relative to the components of $\mathcal{C}$.

Given an observed change in $\mathcal{C}[M \mid D]$, define a correction operator Φ acting on methods:

$$M_{t+1} = \Phi(M_t, D, \mathcal{R}_t),$$

where $\mathcal{R}_t$ is the history of local, composite, and global test outcomes up to time t .

Corrective adjustment may involve revising method scope, adjusting intensity of application, revising uncertainty bounds, or refining internal configuration based on observed test outcomes.

A correction cycle is effective when

$$\mathcal{C}[M_{t+1} \mid D] \geq \mathcal{C}[M_t \mid D],$$

and local alignment, veracity, and uncertainty conditions and composite test conditions are satisfied over a finite horizon.

B.8 Testing Protocol (Summary)

Given a method M and domain D , testing proceeds by specifying the method's purpose and constructing the set of evaluative interfaces. Local tests assess alignment, veracity, and uncertainty on selected interfaces, followed by composite tests examining interaction among components. Global alignment is then evaluated using $\mathcal{C}[M \mid D]$. Diagnostic patterns may prompt corrective adjustment through Φ , after which testing may be repeated to examine generative continuation.

B.9 Comparative Examples

The following examples function as controlled calibration cases. Their scope is deliberately limited and intended merely to illustrate the testing mechanics.

Each example fixes in advance a domain of application, a reference configuration, a set of evaluative interfaces, alignment, veracity, and uncertainty measures, together with weighting parameters and thresholds for the global functional $\mathcal{C}$.

Once these elements are specified, the evaluation results follow entirely from the internal mechanics of the testing. The examples are reference instances. They show that the same evaluative criteria register both successful continuation and breakdown.

In ordinary inquiry, methods are rarely organized in this way. Their components interact unevenly and respond to shifting conditions. When Appendix B is applied, evaluation becomes a way of tracing alignment, uncertainty, and continuation.

Common Testing Setup

In both examples below, the testing regime is fixed as follows.

- **Domain D :** a two-dimensional Euclidean plane.
- **Evaluative interface e :** a vertex at which tile edges meet.
- **Reference behavior y_e^{ref} :** the geometric requirement for a complete surface, namely that the sum of internal angles meeting at any vertex equals 360° .
- **Alignment map:**

$$A(\theta, \theta_{\text{ref}}) = 1 - \frac{|\theta - \theta_{\text{ref}}|}{360^\circ}.$$

- **Veracity and uncertainty conventions:**
 - $v_e(M) = 1$ when the reference behavior is met at e ,
 - $u_e(M) = 0$ when the configuration is gapless and repeatable at e .
- **Thresholds:** $a_{\min} < 1$, $v_{\min} < 1$, $u_{\max} > 0$, with specific numerical values fixed as part of the testing regime in the examples below.

The same evaluative interfaces and thresholds are used in both cases.

Example 1 — Collapse: Regular Pentagon Tiling

Let M_{pent} be the method consisting of placing regular pentagons according to a fixed adjacency rule.

A regular pentagon has internal angle 108° . At a vertex where three pentagons meet,

$$y_e(M_{\text{pent}}) = 3 \times 108^\circ = 324^\circ.$$

Applying the alignment map,

$$a_e(M_{\text{pent}}) = A(324^\circ, 360^\circ) = 1 - \frac{36^\circ}{360^\circ} = 0.9.$$

For a testing regime with $a_{\min} = 0.99$, the alignment condition $a_e(M) \geq a_{\min}$ fails, and

$$T_{\text{loc}}(M_{\text{pent}}, D, e) = 0.$$

This failure is systematic. Because the angular relations are intrinsic to regular pentagons, every evaluative interface in $\mathcal{E}_{\text{local}}(M_{\text{pent}}, D)$ exhibits the same misalignment. As the method is extended over the domain, gaps accumulate, preventing a repeatable surface configuration. The veracity condition and the uncertainty bound cease to be satisfied at those interfaces.

Accordingly, the global alignment functional $\mathcal{C}[M_{\text{pent}} \mid D]$ decreases as evaluation interfaces accumulate. When $\mathcal{C}[M_{\text{pent}} \mid D] < \tau_{\text{low}}$, the method enters the **Collapse** diagnostic regime: local alignment, veracity, and uncertainty conditions fail systematically, and the resulting breakdown of compatibility under accumulation causes the composite test conditions to cease to be satisfied in a significant portion of evaluation interfaces, rendering the method not generatively valid.

Example 2 — Generative Validity: Regular Hexagon Tiling

Let M_{hex} be the method consisting of placing regular hexagons according to the standard adjacency rule that produces a uniform hexagonal tessellation.

A regular hexagon has internal angle 120° . At every vertex where three hexagons meet,

$$y_e(M_{\text{hex}}) = 3 \times 120^\circ = 360^\circ.$$

Applying the same alignment map,

$$a_e(M_{\text{hex}}) = A(360^\circ, 360^\circ) = 1.$$

At every evaluative interface,

$$v_e(M_{\text{hex}}) = 1, \quad u_e(M_{\text{hex}}) = 0,$$

and all local alignment, veracity, and uncertainty conditions are satisfied:

$$T_{\text{loc}}(M_{\text{hex}}, D, e) = 1 \quad \forall e \in \mathcal{E}_{\text{local}}(M_{\text{hex}}, D).$$

Because the method operates uniformly at a single scale and introduces no interaction mismatch, composite test conditions are also satisfied:

$$T_{\text{comp}}(M_{\text{hex}}, D) = 1.$$

The global alignment functional evaluates to

$$\mathcal{C}[M_{\text{hex}} \mid D] = \lambda_A \cdot 1 + \lambda_V \cdot 1 - \lambda_U \cdot 0 = \lambda_A + \lambda_V,$$

which exceeds τ_{high} for any domain-appropriate choice of weights with $\lambda_A + \lambda_V \geq \tau_{\text{high}}$.

Accordingly, M_{hex} is generatively valid on D under the specified testing regime.

Comparative Interpretation

Appendix B distinguishes failure from compatibility using a fixed evaluative interface, a shared reference behavior, and a single alignment map. The regular pentagon method fails because its intrinsic geometric relations force persistent failure of the local alignment condition at evaluative interfaces, driving $\mathcal{C}[M_{\text{pent}} \mid D]$ below τ_{low} . The regular hexagon method satisfies the local alignment, veracity, and uncertainty conditions and the composite test conditions under identical evaluative assumptions, and attains $\mathcal{C}[M_{\text{hex}} \mid D] \geq \tau_{\text{high}}$, showing generative validity. The contrast demonstrates that Appendix B diagnoses whether a method can satisfy local, composite, and global alignment requirements when extended over its domain.

APPENDIX C — SYSTEMIC CASE STUDY

Iterative Fuel Tuning with Closed-Loop Correction

This case study applies the constructions of Annex I (Sheaf Theory), Annex II (Algebraic Composition), and Appendix B (Testing) to a well-known mechanical system. It demonstrates how algebraic incompatibility between update operators and restriction maps leads to persistent failure in iterative methods.

C.1 System and Context

Consider a modern fuel-injected motorcycle managed by an Engine Control Unit (ECU). The primary function of the ECU is to determine the precise extent of the fuel injection pulse required for each engine cycle.

The system relies on a Base Fuel Map: a static lookup table indexed by engine speed (RPM) and load (e.g., throttle position or intake manifold pressure). This map provides a baseline value for the injector pulse width.

To compensate for inaccuracies in the Base Map, the system utilizes Closed-Loop Correction. Oxygen sensors in the exhaust provide feedback on the actual Air-Fuel Ratio (AFR). If the measured value differs from the target AFR, the system applies a real-time adjustment to the fuel delivery.

The specific configuration under study includes a separate, programmable controller capable of modifying the ECU's behavior. This controller implements an automated tuning workflow consisting of two distinct operations:

- **Trim Accumulation:** The controller monitors the closed-loop corrections applied during operation. It stores these

correction values in a temporary memory buffer. These stored values are termed ‘trims’ (additive offsets required to align the base map with the measured fueling requirement).

- **Map Update (‘Accept Trims’) Φ :** A procedure that permanently adds the stored trim values to the Base Fuel Map and resets the temporary buffer to zero.

The purpose of this workflow is to refine the Base Fuel Map so that it becomes accurate enough to operate with minimal reliance on feedback.

The engine operates over a domain D partitioned into regimes such as steady-state cruise, transient low-load operation, high-load/wide-open throttle, and deceleration.

Closed-loop correction is active only on a proper subdomain $D_c \subset D$ (typically steady-state low-to-mid load operation). Outside this region (D_o), AFR feedback is ignored, and the system relies solely on the Base Fuel Map (open-loop operation).

The method system under study is this combined tuning workflow: a model of iterative adaptive feedforward learning under partial observability.

C.2 Constituent Methods

The system involves the following interacting methods.

1. **Base Fuel Mapping F :** A table indexed by RPM and load specifying **injector pulse widths** (the duration of time the fuel injector remains open). It operates open-loop.
2. **Closed-Loop Correction C :** A feedback mechanism that adjusts fueling in real-time based on oxygen sensor measurements to achieve the Target AFR. Active only on D_c .

3. **Trim Accumulation T :** A procedure that records persistent deviations between measured and target AFR under $F \otimes C$. Produces a table of trims supported mainly on D_c .
4. **Update Operator ('Accept Trims') Φ :** A procedure that updates the base map by embedding the accumulated trims and resetting the trim table:

$$F \leftarrow F + T, \quad T \leftarrow 0,$$

leaving C unchanged.

5. **Iteration Protocol W :** The documented workflow:
 - (a) Operate the motorcycle under automation.
 - (b) Accumulate trims.
 - (c) Accept trims.
 - (d) Repeat.

This composite protocol is the object of evaluation.

C.3 Domain Organization

Let:

$$\begin{aligned} D &= \text{full operating domain,} \\ D_c &\subset D = \text{closed-loop domain,} \\ D_o &= D \setminus D_c. \end{aligned}$$

Properties:

- $C \neq 0$ on D_c (Feedback active).
- $C = 0$ on D_o (Feedback inactive).

Trim generation is concentrated on D_c . Updates are applied globally across D .

This produces a mixed-regime update **process**.

C.4 Observed Practical Phenomenon

A recurrent practical outcome is:

After repeated trim–accept cycles,

- logged AFR values appear close to targets,
- average trims become small,
- yet the motorcycle exhibits **steady-throttle surging** (instability/oscillation), poor throttle response, and thermal irregularities.

Further iterations frequently fail to resolve these symptoms. Practitioners describe this as ‘the ECU fighting the tune.’

C.5 Formal Representation

Model composite fueling behavior as:

$$M = F \otimes C.$$

Define restriction:

$$R_{D_c}(M) = F|_{D_c} \otimes C|_{D_c}, \quad R_{D_o}(M) = F|_{D_o}.$$

Define closed-loop trim generation:

$$T_c := \text{trim}_c(F, C),$$

with T_c supported mainly on D_c .

Define open-loop trim generation (if present):

$$T_o := \text{trim}_o(F|_{D_o}),$$

typically weak, noisy, or absent.

Define the implemented update operator:

$$\Phi(F, C) := (F + T_c, C).$$

C.6 Annex II Result: Non-Commuting Update and Restriction

Annex II requires compatibility of update with restriction.

Consider restriction to D_o .

Restrict after update:

$$R_{D_o}(\Phi(F, C)) = (F + T_c)|_{D_o}.$$

Update after restriction:

Define the open-loop update operator

$$\Phi_o(F|_{D_o}) := F|_{D_o} + T_o.$$

Then:

$$\Phi_o(R_{D_o}(F, C)) = F|_{D_o} + T_o.$$

Because T_c is learned under closed-loop feedback semantics and T_o is generated under open-loop semantics,

$$(F + T_c)|_{D_o} \neq F|_{D_o} + T_o.$$

Therefore,

$$R_{D_o}(\Phi(F, C)) \neq \Phi_o(R_{D_o}(F, C)).$$

The update operator does not commute with restriction.

This incompatibility is intrinsic to the workflow.

C.7 Operational Consequence

Because update and restriction do not commute:

1. Corrections are learned in a feedback-mediated regime.
2. They are embedded into a map used in a non-feedback regime.

3. On return to D_c , feedback re-enters and re-corrects.
4. New trims are generated.
5. The cycle repeats.

The iteration

$$(F, C) \mapsto \Phi(F, C) \mapsto \Phi^2(F, C) \mapsto \dots$$

does not admit a stable fixed point on trajectories that cross $D_c \leftrightarrow D_o$.

Local convergence on D_c can coexist with global non-convergence in mixed-regime trajectories.

C.8 Annex I Consequence: Failure of Global Gluing

On each regime:

- On D_c , the behavior of $F \otimes C$ stabilizes locally.
- On D_o , the behavior of F stabilizes locally.

However, the update exports corrections to the regime boundary in a manner incompatible with both restricted behaviors.

The locally stable sections on D_c and D_o do not assemble into a single global section invariant under iteration.

The descent/gluing condition fails.

Local correctness does not entail global coherence.

C.9 Appendix B Classification: Evaluation-Set Instability

Define evaluation sets:

- $E_1 \subset D_c$: steady-state sampled cells,

- E_2 : cross-regime trajectories and long-term operation.

Under E_1 :

- sampled AFR values align with targets,
- trims remain small,
- local veracity conditions are satisfied.

Under E_2 :

- oscillation persists,
- uncertainty accumulates,
- continuation fails.

Accordingly,

$$\mathcal{C}[M|E_1] \geq \tau_{\text{high}}, \quad \mathcal{C}[M|E_2] < \tau_{\text{low}}.$$

The workflow overfits restricted evaluation and fails under broadened evaluation.

C.10 Forced Remedies

The analysis yields two workable **strategies**.

Strategy A: Domain-Respecting Update Redefine Φ so that:

$$F|_{D_o} \text{ is not updated using } T_c.$$

Accept trims only within their generating domain, or maintain separate regime-specific maps.

This restores update–restriction compatibility.

Strategy B: Remove Feedback Disable C .

Then:

$$M = F, \quad \Phi(F) = F + T_o,$$

and restriction trivially commutes.

Hybrid strategies remain unstable.

C.11 Added Diagnostic Content

Practitioner explanation: ‘Closed loop fights the tune.’

Systemic diagnosis: The documented trim–accept workflow violates compositional compatibility between update and restriction. Iteration is therefore **inherently** non-convergent on mixed domains. No amount of parameter adjustment or repetition can produce convergence without modifying the update **logic**.

C.12 Register Validity

This case satisfies the criteria for a systems-of-methods example:

- multiple interacting methods,
- layered composition,
- explicit domain partition,
- iterative refinement,
- persistent failure mode,
- documented workflow,
- independence from individual tuning skill.

The result follows mechanically from the interaction of operators.

